Laboratory: GNSS Positioning with the SPP Method

Week 9 — Collect, validate, compete on a leaderboard

AAE4203 – Guidance and Navigation

Dr Weisong Wen Assistant Professor

- **Lab** Trustworthy AI and Autonomous Systems Laboratory (TAS Lab)
- **Dept** Aeronautical and Aviation Engineering, PolyU
- **Time** Week 9, Semester 1, 2026/27 · 30 October 2026

7	57	20	1
sections	slides	groups	leaderboard

AAE4203 · Semester 1, 2026–27 · PolyU

THE BIG PICTURE



By the end of the lab you can:

- collect raw GNSS data with u-center and a ZED-F9P (15–20 min per group)
- process and validate your own data with RTKLIB and your Python SPP code
- improve accuracy with AI-assisted coding: multi-constellation, weighting, outlier rejection
- submit a benchmark trajectory and compete on the leaderboard

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Plan, field collection, processing and validation, the benchmark challenge, the report and a worked example

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Session plan, the three datasets and the groups

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Software, receiver set-up and u-center logging

03

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Tutorial, manuals and references

Lab Introduction



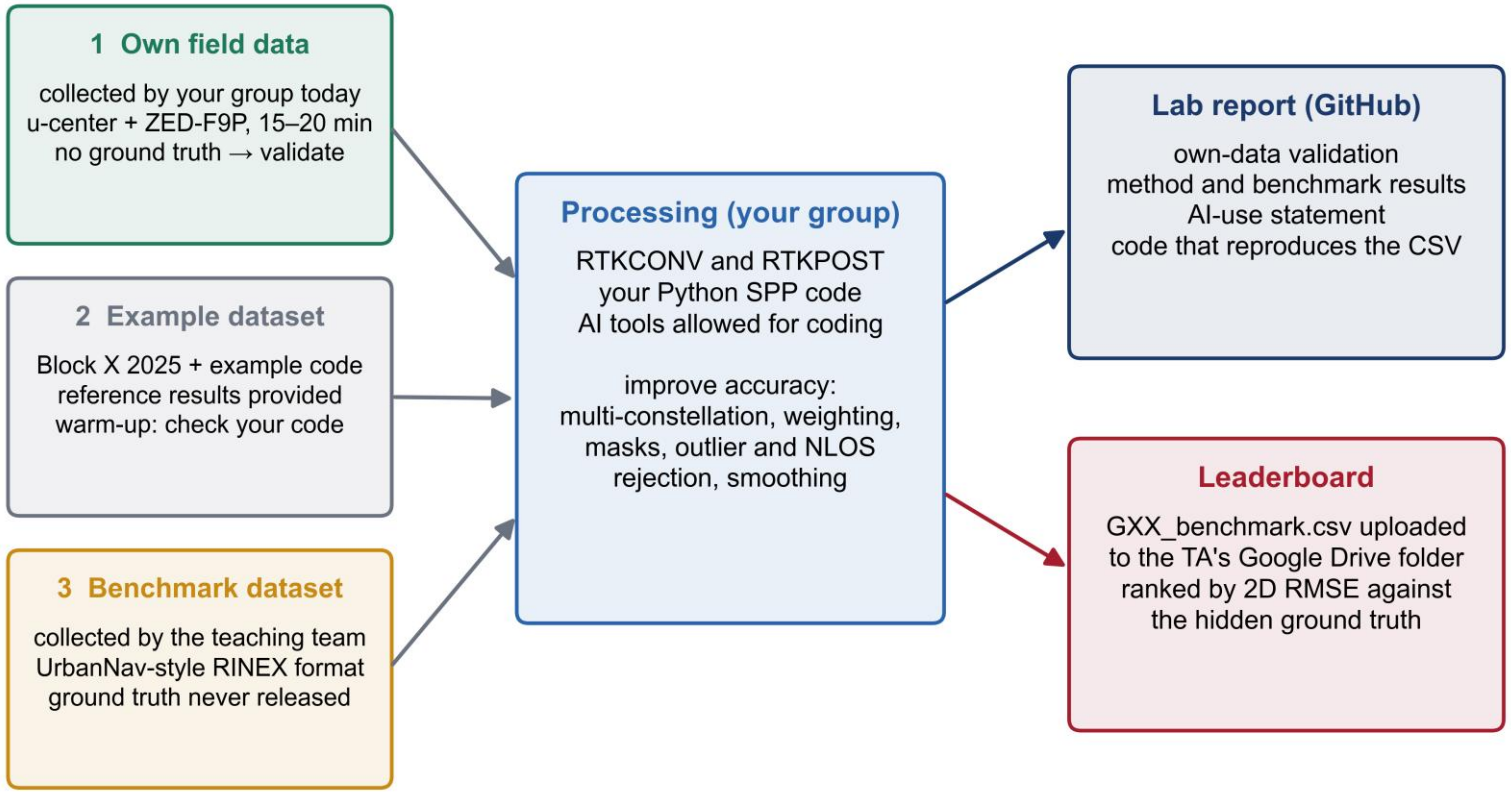


This lab focuses on the practical application of GNSS, from data collection with a receiver to data processing using both professional software and custom Python scripts. Upon successful completion of this lab, you will be able to:

- ***Objective 1***: Operate GNSS data collection software, especially u-center for u-blox receivers.
- ***Objective 2***: Process raw GNSS data using the open-source RTKLIB software suite to estimate position and velocity.
- ***Objective 3***: Develop Python scripts to process GNSS data for position estimation.
- ***Objective 4***: Analyze GNSS data, including the evaluation of positioning error for a static dataset collected in a challenging urban environment, using Python.

The Lab at a Glance (2026/27)

Three datasets, one processing chain, two outputs



What is new this year

Own data — every group collects and validates its own measurements

Benchmark — urban dataset from UrbanNav

Leaderboard — upload a CSV trajectory; groups are ranked by accuracy

AI tools — allowed for coding, with rules

5 kits, 4 batches — same steps; batch 2-4 starts with the example

Collect → validate your own data → compete on the benchmark

01

Lab Plan & Groups

Section 1 of 7

- Session plan: 5 kits, 4 batches
- Three datasets, three questions
- Groups G01–G20

1 **Lab Plan & Groups**

2 Field Data Collection

3 Processing & Own-Data Validation

4 Benchmark Challenge & Leaderboard

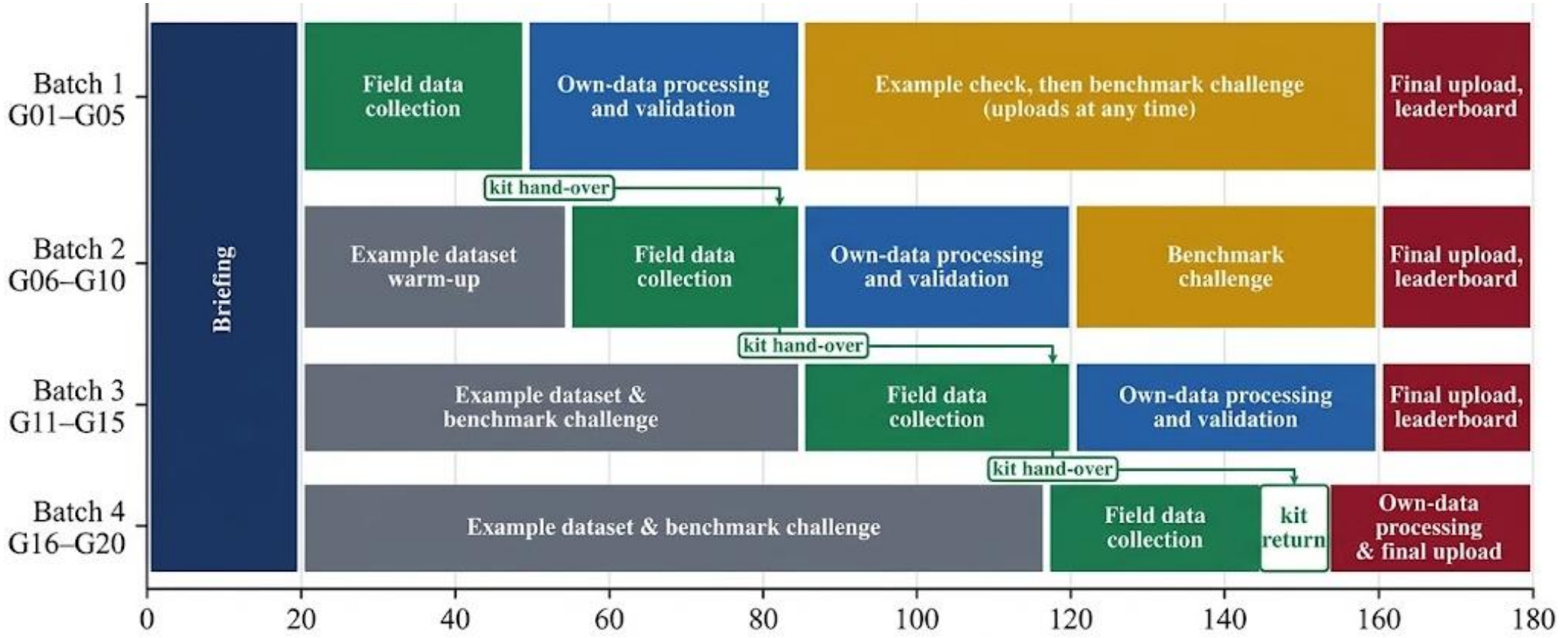
5 Report & Checklist

6 Example Dataset: Block X

7 Resources

Session Plan: 5 Kits, 4 Batches

3-hour session on 30 October 2026 · every group: collect → validate own data → benchmark



Final ranking uses the files at the deadline, so all 4 batches have the same total time

Three Datasets, Three Questions

Each dataset answers a different question about your work

1 Your own data

Can we collect and process real data?

What: static, 15–20 min, near Block X

Truth: none → validate by comparison

When: 4 batches (0:20, 0:55, 1:30, 2:05)

Output: own-data part of the report

2 Example dataset

Does our code give the right answer?

What: Block X, 2025, UBX/RINEX/CSV + example code

Truth: reference results provided

When: before the lab (batch 2 also at 0:20) and after every code change

Output: confidence that the code works

3 Benchmark dataset

How accurate is our method?

What: outdoor data in Hong Kong, UrbanNav-style RINEX

Truth: kept by the teaching team for class evaluation

When: after validating your own data

Output: leaderboard rank

Only the benchmark is ranked — a hidden ground truth makes the comparison fair

Grouping (2026/27)

Same groups as the group presentation · one kit, one dataset and one report per group

78 students → 20 groups → 18 × 4 and 2 × 3 → 5 kits · 4 batches

Each group	Suggested roles
Borrows one kit: u-blox ZED-F9P + antenna (bring a laptop) Collects one dataset near Block X Submits one benchmark CSV, one own-data CSV and one report	1 Hardware set-up and logging (u-center) 2 Conversion and RTKLIB processing 3 SPP coding and accuracy improvement 4 Analysis, report and GitHub repository

Every member must be able to explain the whole workflow and the submitted code

Group List: G01–G10

From AAE4203_Group_List_v7.csv · members in alphabetical order

Group	Batch	Kit	Members
G01	**1**	K01	CHEN Che, HAU Kin Chung, WANG Ki Tai, YANG Chenyu
G02	**1**	K02	JIANG Runqian, LEE Hon Yan, LEUNG Chi Kin, XIAO Suen
G03	**1**	K03	NG Tsz Hin, SELVY Can, SUN Yinlan, YANG Yuqi
G04	**1**	K04	LI Jiasheng, WANG Zicheng, XU Yinhao
G05	**1**	K05	GUO Feiyang, HE Yinxi, YE Tin Ming, ZHANG Jiaming
G06	**2**	K01	KAO Ching Yiu, LI Yijie, TAM Wing Him, YE Zixuan
G07	**2**	K02	FU Chuen Yik, LIMBU Tsz Ming Amrit, LIN Yan-cheng, ZHOU Yulin
G08	**2**	K03	LIU Boyu, TANG Tsung Wang, YEUNG Hiu Chun, ZHANG Wenxi
G09	**2**	K04	CHAN Man Ho, LEUNG Lok Hang Max, ZHANG Shuxuan
G10	**2**	K05	CHEN Yulin, CHENG Pak Hin, TONG Chi Yip, TSANG Shing Chi

Batch 1 (G01–G05): collect at 0:20–0:50 · Batch 2 (G06–G10): collect at 0:55–1:25

Group List: G11–G20

From AAE4203_Group_List_v7.csv · members in alphabetical order

Group	Batch	Kit	Members
G11	3	K01	DENG Jiabao, LAI Ki Uen, SONG Yujie, WANG Jinpeng
G12	3	K02	CHAN Kai Yin, CHENG Man Wai, LEE Wang Hei, LIN Ka Kin
G13	3	K03	LIANG Kam Sang, LU Jin, LUI King Shing, TANG Zijia
G14	3	K04	SHEK Pui Ka, SUN Jiajun, YANG Jie, ZHOU Jianhong
G15	3	K05	LO Trevor Che Kit, NI Jingnian, WANG Duanqinrui, ZHANG Dingzhe
G16	4	K01	OU Yeyang, SHI Hengxin, SIT Orison, WU Xinyue
G17	4	K02	LEE Tsz Nam, SAMSONOV Aleksandr, TURK Isa, YU Qinxiang
G18	4	K03	CHIU Sung Lim, RAMOS CRUZ Axel Daniel, YU Mengzhi, ZHANG Borui
G19	4	K04	CHOY Tsz Him, LI Xiao, XI Yilin, ZHOU Lianyi
G20	4	K05	CHAI Yuan, SHU Yunxiang, TANG Junxian, ZHANG Junming

Batch 3 (G11–G15): collect at 1:30–2:00 · Batch 4 (G16–G20): collect at 2:05–2:35

02

Field Data Collection

Section 2 of 7

- Step-by-step overview
- Software and code
- Steps 1–3: connect and record

1 Lab Plan & Groups

2 **Field Data Collection**

3 Processing & Own-Data Validation

4 Benchmark Challenge & Leaderboard

5 Report & Checklist

6 Example Dataset: Block X

7 Resources

Field Data Collection: Step by Step

30-minute slot: batch 1 at 0:20, batch 2 at 0:55, batch 3 at 1:30, batch 4 at 2:05 · the next slides show each step

Step	What to do	Time
1	Pick up the kit; connect antenna → ZED-F9P → USB → laptop	5 min
2	Place the antenna at your site near Block X, clear view of the sky	3 min
3	u-center: choose the COM port, check satellites and C/N_0	2 min
4	Record raw data to a .ubx file — keep the antenna still	15–20 min
5	Stop, copy the file, take photos; hand the kit to your partner group	3 min

Good practice

- Note the start and end time
- Do not stand or walk next to the antenna
- Check that the .ubx file keeps growing
- Keep the laptop on mains or a full battery
- Back up the .ubx file twice
- Record weather and surroundings

A still antenna and a complete .ubx file are all you need for the rest of the lab

Software Requirements (1/2)

Software Requirements

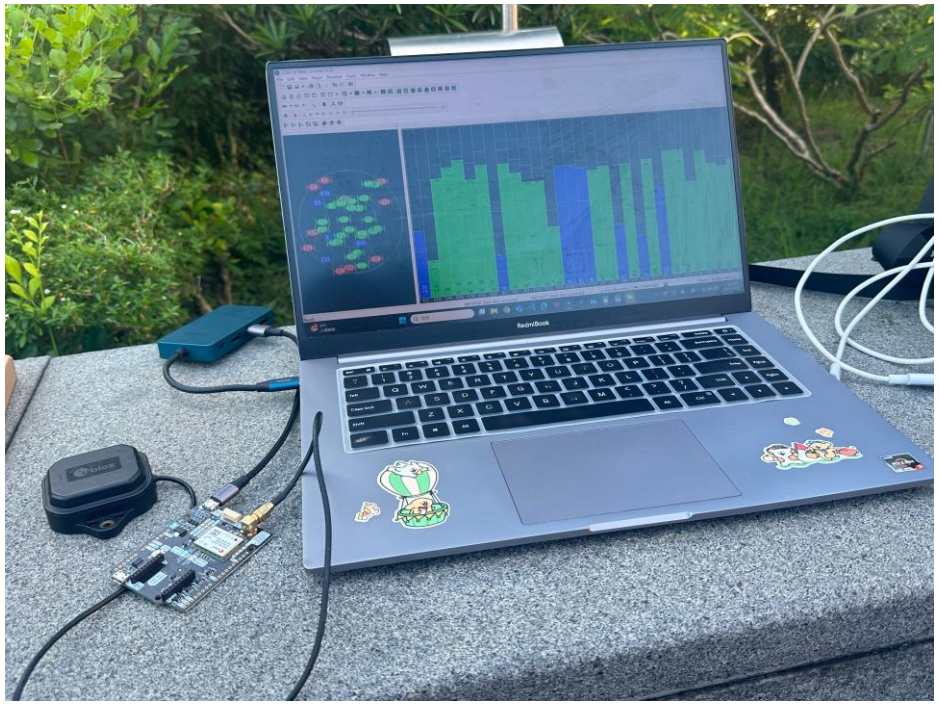
- [u-center](#) for data collection
- [MATLAB](#) with the following toolboxes:
  - Mapping Toolbox
- [Python 3.7+](#) with the following packages:
  - numpy
  - matplotlib
  - pandas
  - pyubx2
  - cartopy (optional, for enhanced map visualization)
- [RTKLIB 2.5](#) (RTKconv and RTKPOST utilities)
- [Google Earth](#) for .kml trajectory visualization

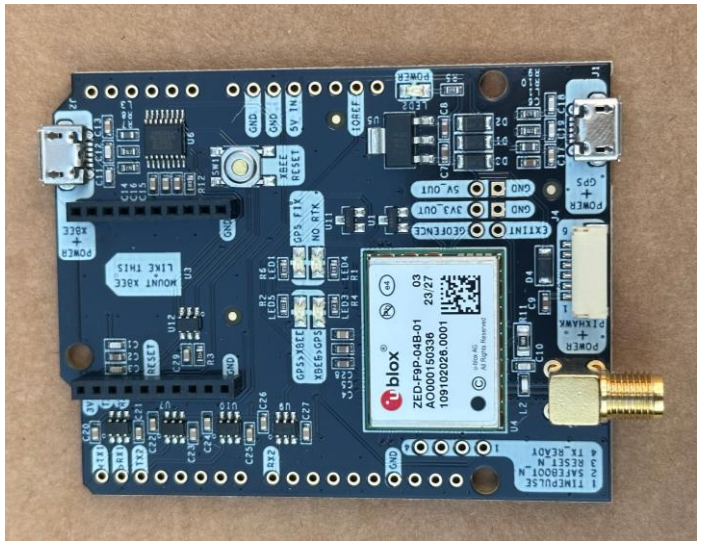


Software Requirements (2/2)

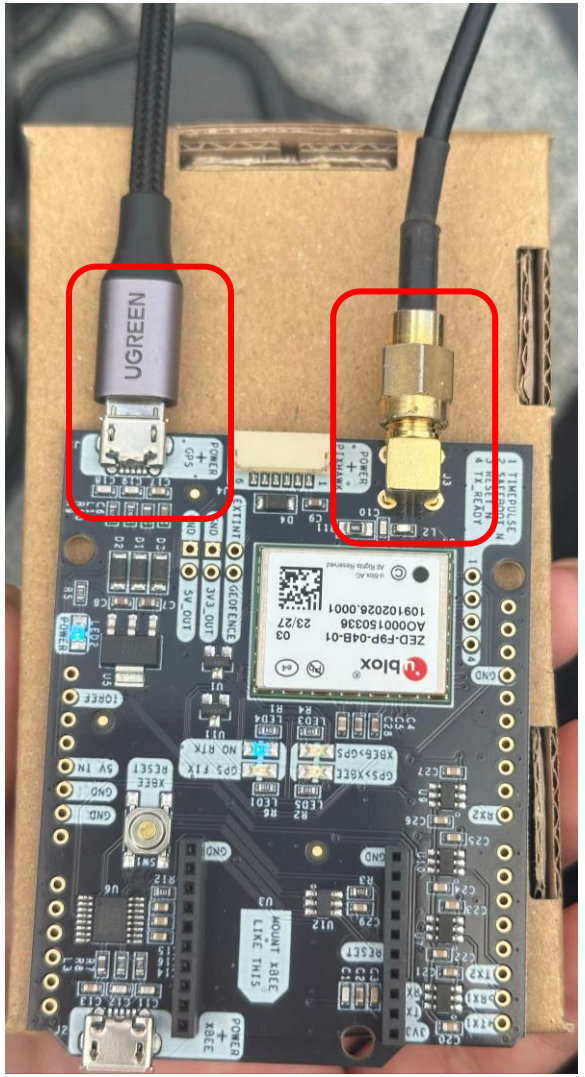
Category	Component	Details
Code	rinex2csv.m	MATLAB script to convert RINEX observation/navigation files into CSV; depends on matRTKLIB.
Code	pyubx2_csv_converter_gui.py	GUI tool to convert UBX/NMEA logs to CSV; depends on pyubx2.
Code	WLSE_spp.py	Reference Python script for Weighted Least Squares Single-Point Positioning ***(provided with the example dataset)***.
Software	u-center	Data collection from u-blox receivers.
Software	MATLAB	Requires Mapping Toolbox.
Software	Python 3.7+	Packages: numpy; matplotlib; pandas; pyubx2; cartopy (optional, for enhanced map visualization).
Software	RTKLIB 2.5	Use RTKconv and RTKPOST utilities for RINEX conversion and post-processing.
Software	Google Earth	Visualize trajectories in .kml format.

Step 1 · Hardware Set-up









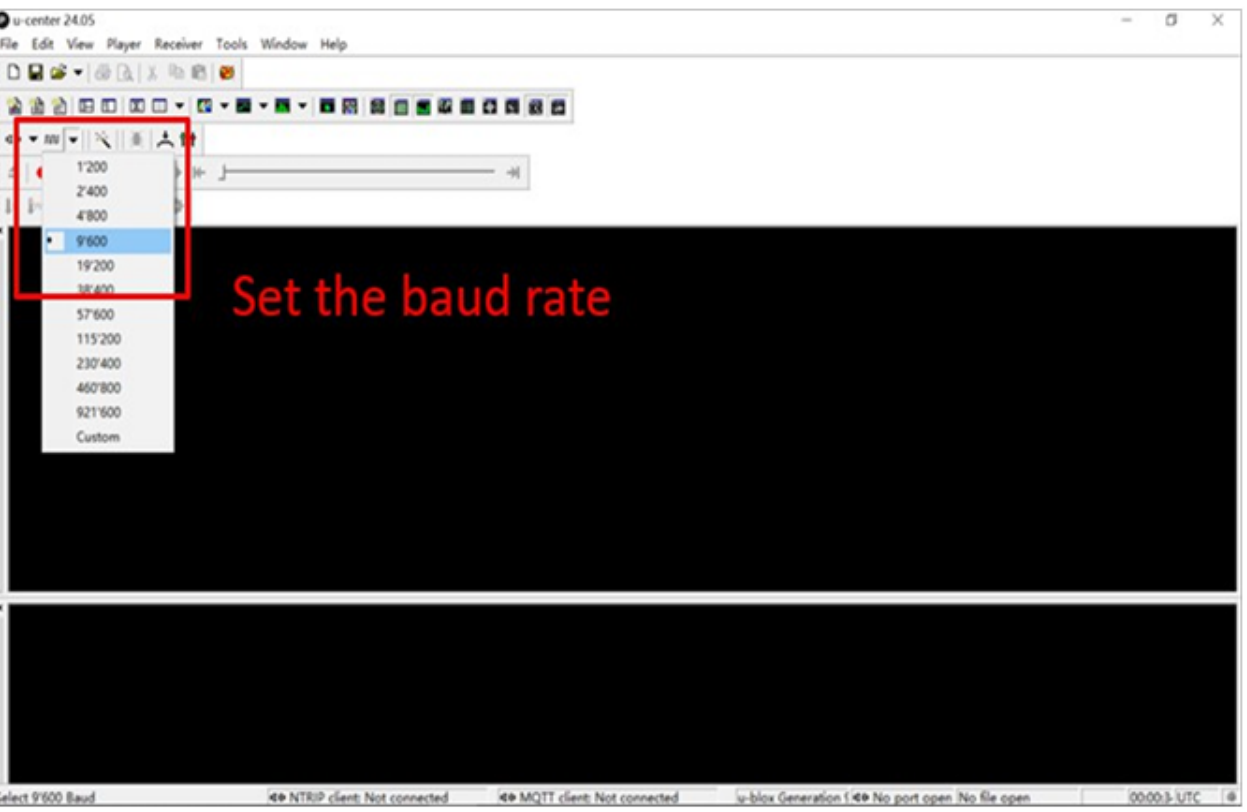
The equipment used for data collection:

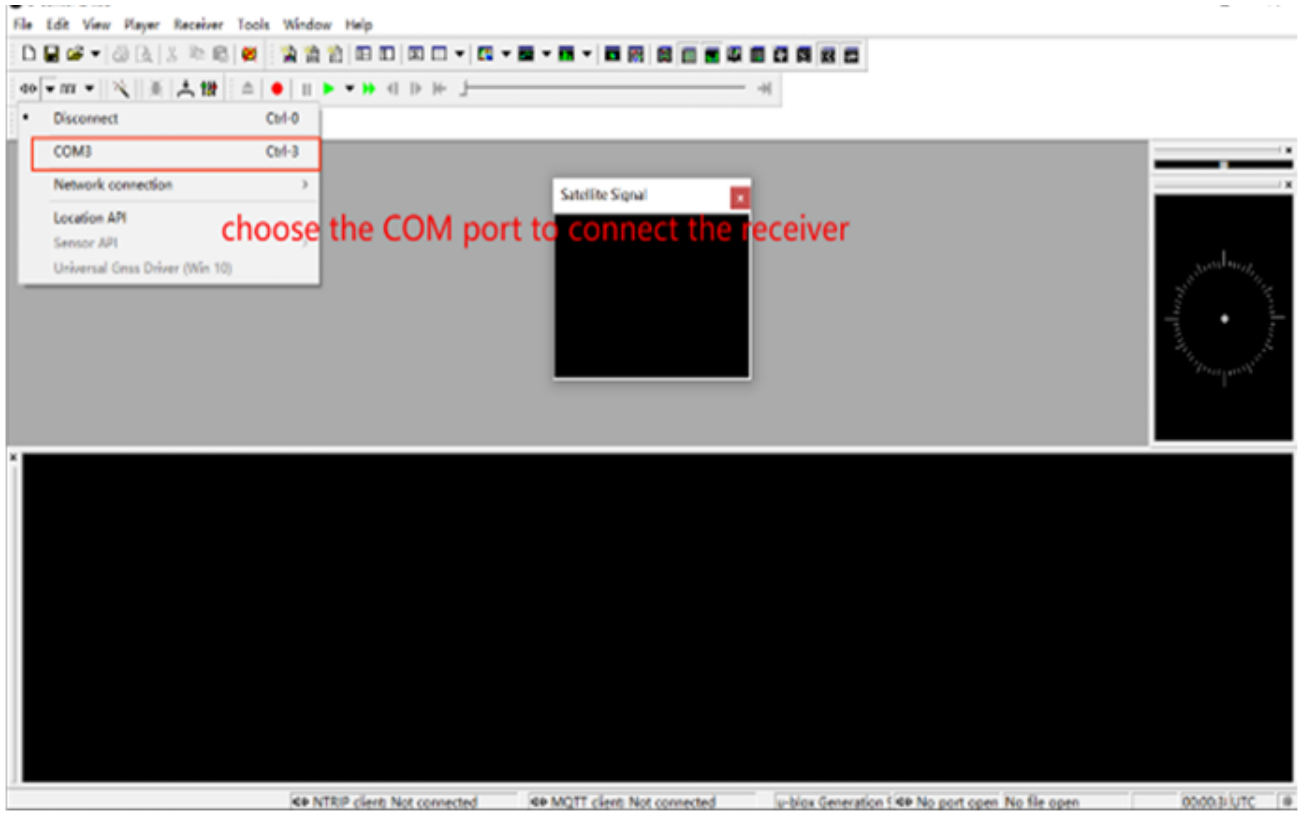
- One laptop
- U-blox ZED-F9P + Antenna / U-blox boards

Device connection details

Step 2 · Connect the Receiver in u-center

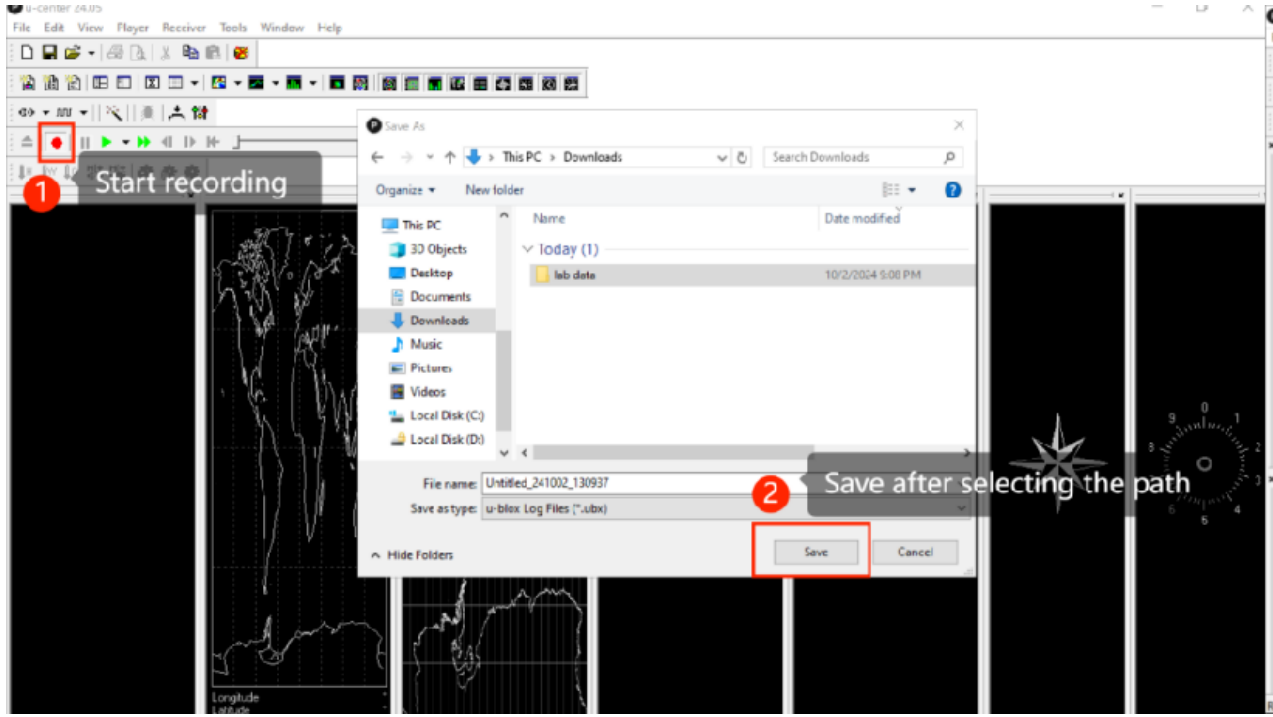
Connect Receiver

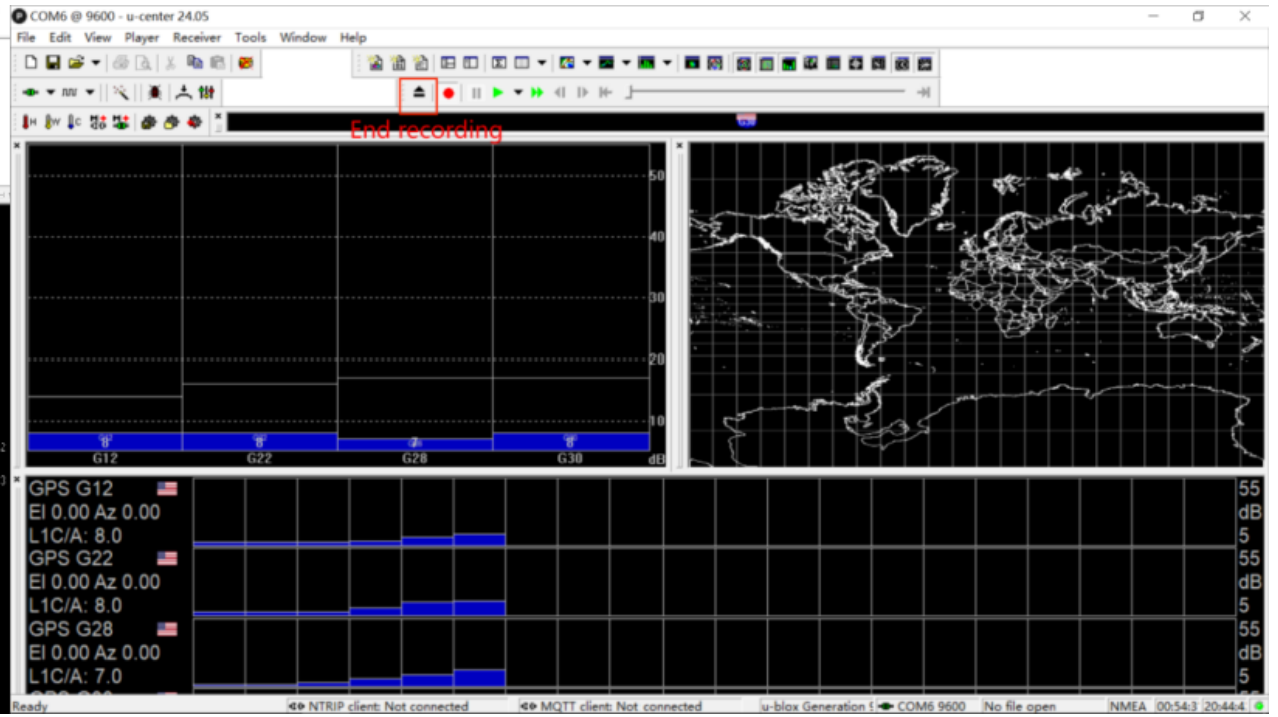




Step 3 · Record Raw Data (UBX)

Data Collection





03

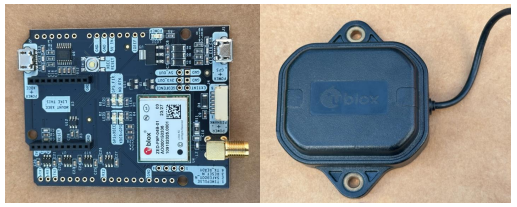
Processing & Own-Data Validation

Section 3 of 7

- Processing pipeline
- Steps 4–8: RTKLIB and CSV files
- Step 9: your SPP code
- Validation without ground truth

1 Lab Plan & Groups
2 Field Data Collection
3 **Processing & Own-Data Validation**
4 Benchmark Challenge & Leaderboard
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Processing Pipeline: Files and Tools





GNSS data collection with u-blox

RTKCONV

.obs .nav

rinex2csv.m

- pseudoranges_meas.csv
- satellite_clock_bias.csv
- ionospheric_delay.csv
- tropospheric_delay.csv
- satellite_positions.csv

WLSE_SPP.py
(Coding by yourself)

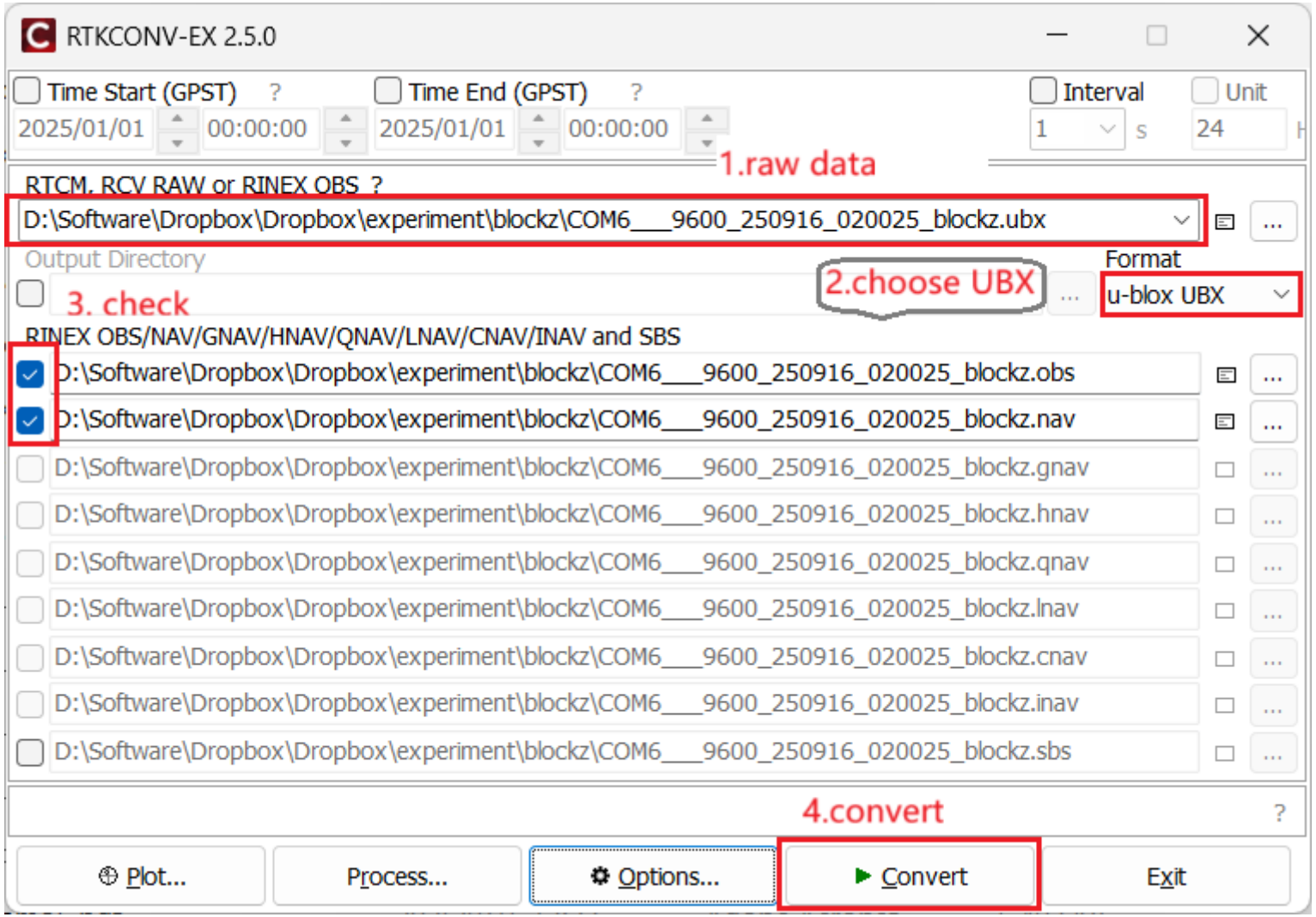
pyubx2_csv_converter_gui.py

NAV-HPPOSECEF.csv
(epoch unit:ms) (position unit: cm)

Receiver reference

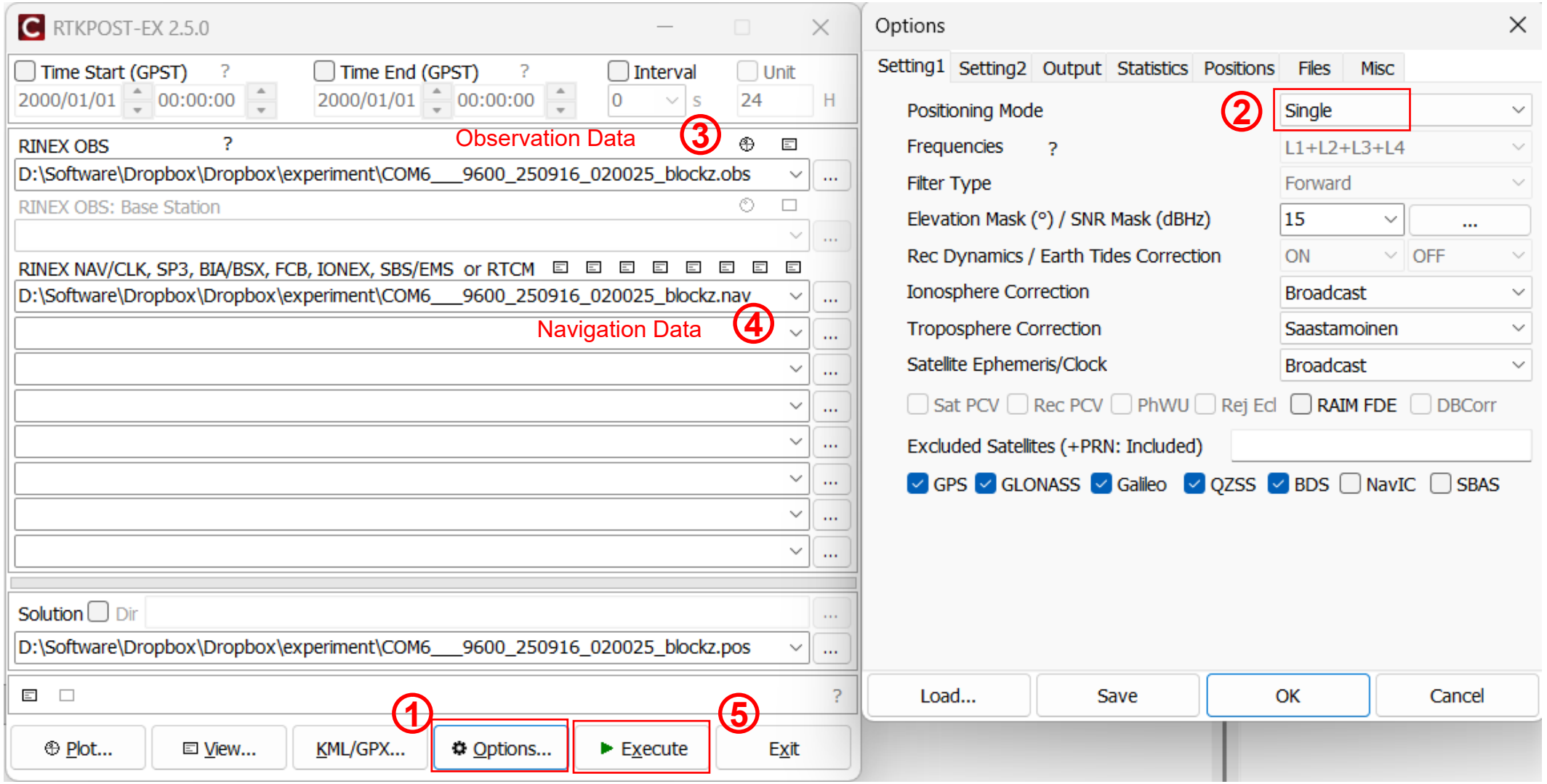
Step 4 · RTKCONV: UBX to RINEX

RTKCONV: UBX → .obs .nav



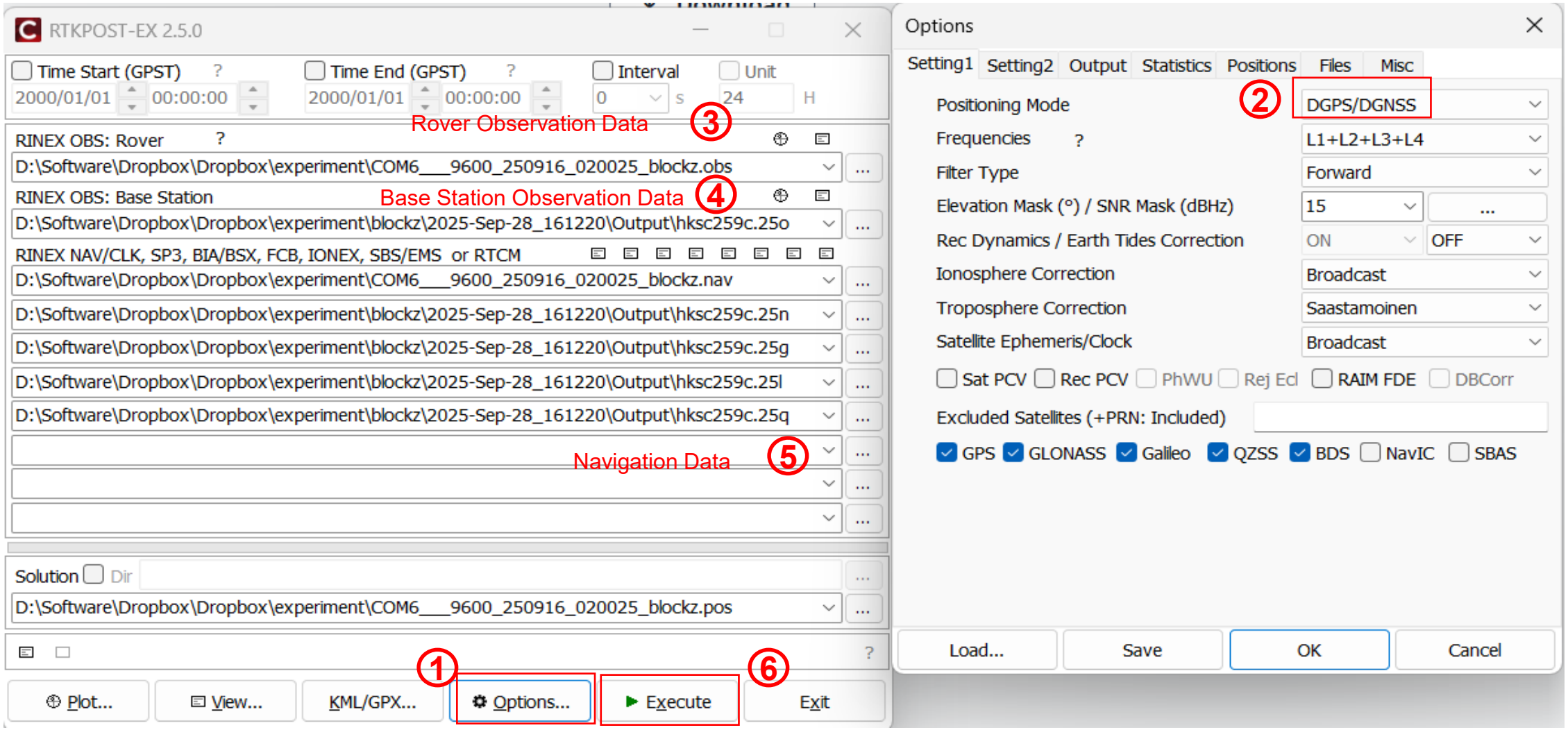
Step 5 · RTKPOST: SPP Solution

RTKPOST: SPP



Step 6 · RTKPOST: DGNSS with HK SatRef

RTKPOST: DGNSS (HK SatRef RINEX)



Step 7 · rinex2csv: RINEX to CSV Inputs

matRTKLIB

```
%% Read RINEX observation/navigation file
obs = gt.Gobs(fullfile(/path/to/your.obs'));
nav = gt.Gnav(fullfile(/path/to/your.nav'));
obs.plot;
obs.plotNSat;
obs.plotSky(nav);
```

```
clc;
clear;
close all;

addpath( fullfile('matRTKLIB') )

%% Read RINEX observation/navigation file
obs = gt.Gobs(fullfile('COM6___9600_250916_020025_blockz.obs'));
nav = gt.Gnav(fullfile('COM6___9600_250916_020025_blockz.nav'));
obs.plot;
obs.plotNSat;
obs.plotSky(nav);

%% Settings for position computation
SNR_TH = 15; % SNR threshold (dBHz)
EL_TH = 10; % Elevation angle threshold (deg)
obs.maskP(obs.L1.S<SNR_TH);
```

```
% Write the matrices to CSV files
writematrix(pseudoranges_meas_mat, 'pseudoranges_meas.csv');
writematrix(satellite_clock_bias_mat, 'satellite_clock_bias.csv');
writematrix(ionospheric_delay_mat, 'ionospheric_delay.csv');
writematrix(tropospheric_delay_mat, 'tropospheric_delay.csv');
writematrix(satellite_positions_mat, 'satellite_positions.csv');
```

COM6___9600_250916_020025_blockz.nav

COM6___9600_250916_020025_blockz.obs



ionospheric_delay.csv

pseudoranges_meas.csv

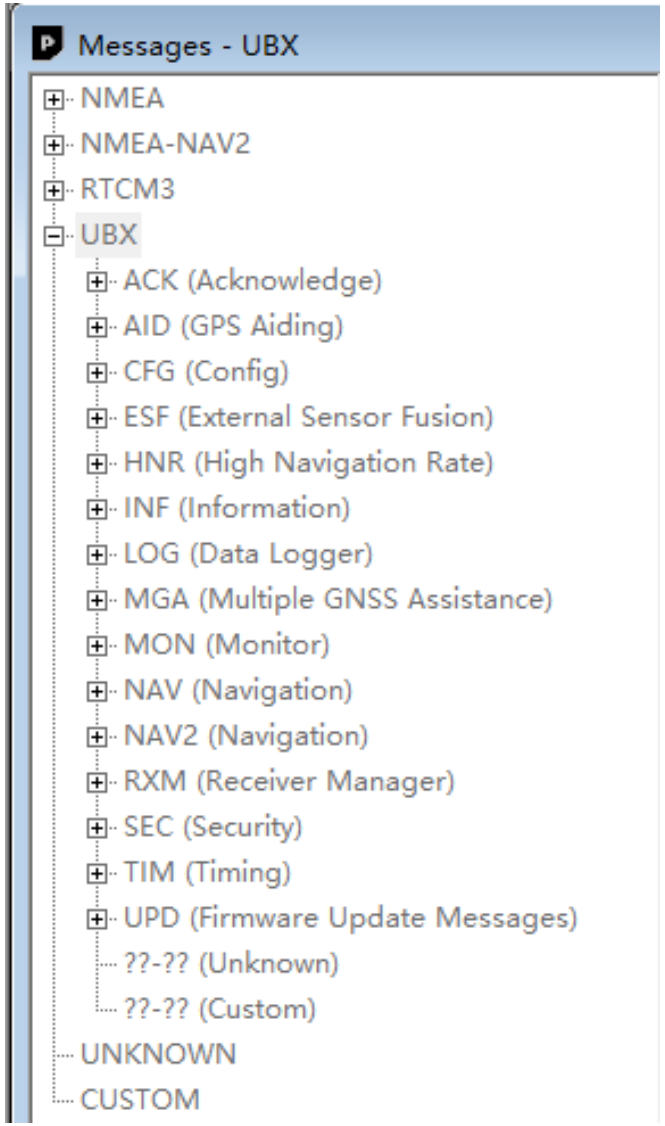
satellite_clock_bias.csv

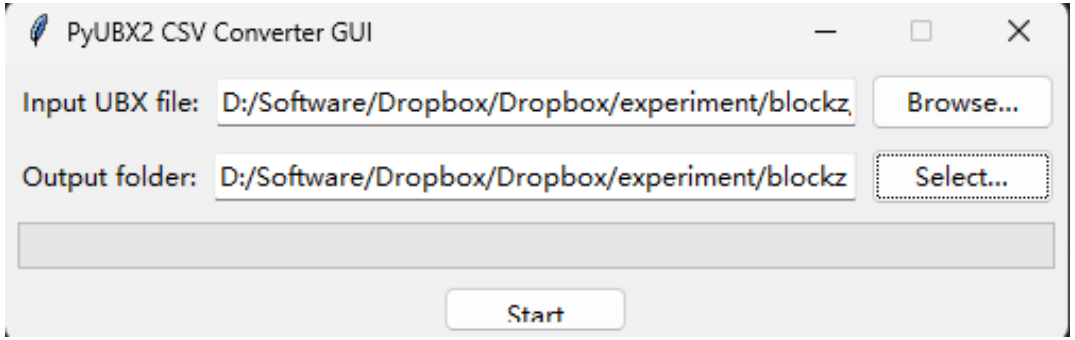
satellite_positions.csv

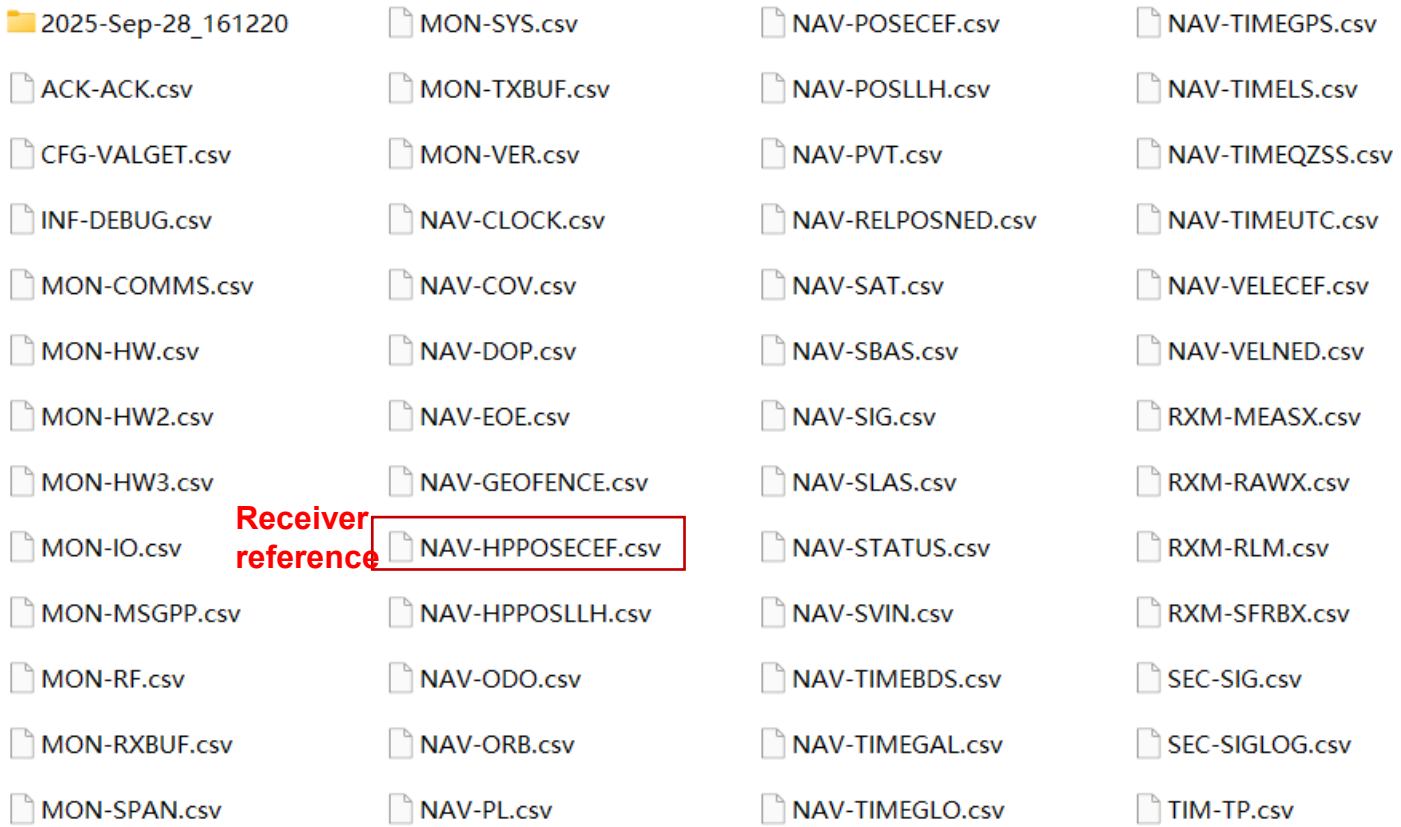
tropospheric_delay.csv

Step 8 · pyubx2: Receiver Solution to CSV

pyubx2_csv_converter_gui.py







Recap: Weighted Least-Squares SPP

What your Python code has to do at every epoch (Weeks 4–6)

$$\rho_i = \|\mathbf{p}^i - \mathbf{p}\| + b - c\,\delta t^i + I_i + T_i + \varepsilon_i$$

$$\mathbf{h}_i^T = [-\mathbf{e}_i^T \;\; 1], \qquad \Delta\mathbf{y} = \mathbf{H}\,\Delta\mathbf{x} + \boldsymbol{\varepsilon}$$

$$\Delta\hat{\mathbf{x}} = (\mathbf{H}^T\mathbf{W}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{W}\,\Delta\mathbf{y}$$

ρ: pseudorange · p: receiver position · b: receiver clock bias (m) · δt^i: satellite clock · I, T: ionosphere, troposphere · e_i: unit line-of-sight vector · W: weights

Algorithm per epoch

1. Read pseudoranges and satellite positions
2. Correct satellite clock, ionosphere, troposphere
3. Start from x_0 (or the last solution)
4. Build Δy and H; solve for Δx
5. Update x; repeat until |Δx| < 1 mm
6. Write the ECEF position to the CSV

Typical bugs

cm vs m, ms vs s (NAV-HPPOSECEF)
wrong sign of the satellite clock term
satellites with missing corrections

Example inputs: the five CSV files produced by rinex2csv.m

Step 9 · Your Task: Least-Squares SPP in Python

Assignment: LSE_SPP Script (Python)

•**Goal**: Implement Single Point Positioning (SPP) using Least Squares Estimation (LSE).

•**Input Files**:

- `pseudoranges_meas.csv`
- `satellite_clock_bias.csv`
- `ionospheric_delay.csv`
- `tropospheric_delay.csv`
- `satellite_positions.csv`

•**Tasks**:

- Read and process the input data.
- Apply corrections (clock bias, ionosphere, troposphere).
- Estimate receiver position with LSE_SPP.
- Output estimated position (ECEF and/or LLH).
- Compare with the receiver solution from u-center (NAV-HPPOSECEF).

Validating Your Own Data Without Ground Truth

Your own data have no reference trajectory — so check them in three ways

Validation asks: is our processing chain working, and how does the solution behave? It does not give the true error.

1 Compare solutions

Your SPP vs RTKPOST SPP

Your SPP vs receiver solution (NAV-HPPOSECEF)

Differences in East, North, Up

2 Check consistency

Static antenna: the scatter is your noise

Mean, standard deviation, 95 % radius

Plot East vs North

3 Check the inputs

Satellites per epoch and per system

C/N_0 and elevation; DOP over time

Post-fit residuals: look for outliers

In the report: one table of mean and standard deviation of the E/N/U differences for the three solutions, and one East–North scatter plot.

Agreement is not accuracy: all three solutions can share the same multipath error

04

Benchmark Challenge & Leaderboard

Section 4 of 7

- The benchmark challenge
- How to improve accuracy
- Using AI tools
- Ranking, CSV format and upload

1 Lab Plan & Groups

2 Field Data Collection

3 Processing & Own-Data Validation

4 **Benchmark Challenge & Leaderboard**

5 Report & Checklist

6 Example Dataset: Block X

7 Resources

The Benchmark Challenge

Same algorithm, a harder dataset, a hidden ground truth

The benchmark dataset	Your task
UrbanNav-HK-Deep-Urban-1, recorded outdoors in Hong Kong	Estimate a position for every epoch
RINEX observation + navigation files (u-blox)	Improve your SPP one step at a time
Reference trajectory from a high-grade system	Use any GNSS technique — only the observation and navigation files as input
Harder than Block X: multipath and blocked signals	Upload GXX_benchmark.csv — as often as you like

Baseline SPP → **One change** → **Check on example** → **Upload CSV** → **See your rank**

Keep a log of every version and its result — it becomes the ablation table of your report

How to Improve Accuracy

Change one thing at a time and measure its effect

Technique	Idea	Expected effect
Multi-constellation	add BDS and Galileo, one clock bias per system	more satellites, better geometry
Weighting	$\sin^2$(elevation) or C/N_0-based variances	down-weights weak, low signals
Masks	elevation 10–15° , C/N_0 25–30 dB-Hz	removes the worst signals
Outlier rejection	drop the largest normalised residual, re-solve	removes multipath / NLOS outliers
Atmosphere	check Klobuchar and Saastamoinen corrections	removes metre-level biases
Filtering	carrier smoothing or a Kalman filter with Doppler	reduces noise

$$\mathbf{x} = [x,\ y,\ z,\ b_G,\ b_C,\ b_E]^T, \qquad \mathbf{h}_i^T = [-\mathbf{e}_i^T\ \ 0\ \ 1\ \ 0] \ \text{(BDS satellite)}$$

Multi-constellation usually gives the largest single gain in Hong Kong

Using AI Tools: Allowed, with Rules

ChatGPT, Claude, Gemini, DeepSeek, Copilot … may be used to write and debug code

Good uses	Rules
Generate a first WLS-SPP from the CSV inputs	Every member can explain the algorithm and the code
Extend it: GPS + BDS + Galileo, weighting, outlier rejection	Verify each version on the example dataset first
Explain error messages and plots	Report: AI-use statement + key prompts in an appendix
Write the evaluation script for the example dataset	The CSV must be computed by your code from the provided observations

Example prompt

“My inputs are pseudoranges_meas.csv, satellite_positions.csv, satellite_clock_bias.csv, ionospheric_delay.csv and tropospheric_delay.csv (columns: …). Write a Python function for weighted least-squares SPP per epoch that returns ECEF positions, and explain each step.”

AI writes code fast; you are responsible for checking that it is right

How You Are Ranked

Benchmark only · errors in the local East-North-Up frame at the reference position

$$\mathrm{RMSE}_{2D} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\left(e_{E,k}^2 + e_{N,k}^2\right)}, \qquad A = 100\,\frac{N}{N_{ref}}\ \%$$

Ranking rules

Ranked only if availability A ≥ 90 %

Smallest 2D RMSE first; ties broken by 3D RMSE

95th-percentile 2D error also shown

Fair play

Your GitHub code must reproduce the CSV

The teaching team may re-run any group's code

Weight of the rank in the lab mark: on Blackboard

Epochs matched by GPS week + time of week · missing epochs lower the availability

Submission File Format (CSV)

One header line + one row per benchmark epoch, in time order

Column	Meaning	Unit / format
gps_week	GPS week number	integer
gps_tow_s	GPS time of week of the epoch	s, at least 1 decimal
x_ecef_m	Estimated receiver position, ECEF X	m, 3 decimals
y_ecef_m	Estimated receiver position, ECEF Y	m, 3 decimals
z_ecef_m	Estimated receiver position, ECEF Z	m, 3 decimals

File names

GXX_benchmark.csv
GXX_owndata.csv
e.g. G05_benchmark.csv

```
gps_week,gps_tow_s,x_ecef_m,y_ecef_m,z_ecef_m
2158,455346.0,-2418202.586,5385847.524,2405648.799
2158,455347.0,-2418202.174,5385847.238,2405648.952
…
```

ECEF in metres (WGS-84) · no extra columns · run python tools/check_submission.py before uploading

Upload and Leaderboard

Google Drive folder from the TA (link on Blackboard), one sub-folder per group

Compute CSV → Run the checker → Upload to Drive → Auto-scoring → Leaderboard

During the session

Upload as soon as you have a first solution

Replace the file whenever you improve it

Live leaderboard link from TA; preliminary ranking at about 2:50

After the session

Keep improving until the deadline (on Blackboard)

The file present at the deadline is final

Final leaderboard on Blackboard; the ground truth stays private

Path: TA's Drive folder / G05 / G05_benchmark.csv (use your own group number; put GXX_owndata.csv in the same sub-folder)

Only the latest file named GXX_benchmark.csv is evaluated

What the Leaderboard Looks Like

Illustrative numbers only — the real leaderboard is shown during the session

Rank	Group	Version	2D RMSE (m)	3D RMSE (m)	2D P95 (m)	Availability
1	G07	GPS+BDS+GAL, C/N_0 weights, outlier rejection	6.8	14.9	12.1	99.4 %
2	G15	GPS+BDS, elevation weights	9.3	18.2	17.5	100 %
3	G02	GPS+BDS, elevation mask 15°	12.6	24.0	23.8	97.1 %
…	…	…	…	…	…	…
18	G11	GPS only (baseline)	31.4	45.7	52.0	100 %
–	G04	GPS only, many epochs missing	8.9	16.3	15.0	71.0 % < 90 %

The last row is not ranked: a low error on a few easy epochs does not count when availability is below 90 %.

Version column: optional GXX_version.txt in your group's Drive folder

05

Report & Checklist

Section 5 of 7

- Lab report requirements
- Checklist

1 Lab Plan & Groups

2 Field Data Collection

3 Processing & Own-Data Validation

4 Benchmark Challenge & Leaderboard

5 **Report & Checklist**

6 Example Dataset: Block X

7 Resources

Lab Report Requirements

One report per group · ≤ 10 pages excluding the cover · code listings ≤ 2 pages

Part	Required content
1. Data collection	set-up, site photos, time, environment near Block X
2. Raw data analysis	satellites per constellation, C/N_0, sky plot, DOP
3. SPP algorithm	model, linearisation, weighted least squares, your code
4. Own-data validation	your SPP vs RTKLIB and the receiver solution; scatter of the static solution
5. Benchmark results	each technique tried, its measured effect, final metrics and rank
6. AI-use statement	tools, purposes, key prompts (appendix), how you verified the output

Template on Blackboard (PolyU_AAE4203_Lab_report_template.docx) · GitHub repository with raw data, code and README · deadline announced on Blackboard

Your repository must reproduce both CSV files

Checklist

Tick everything before you leave the field, the lab, and before the deadline

Before leaving the field

- ☐ UBX file saved and copied to a second place
- ☐ Photos of the set-up and the sky
- ☐ Start/end time, weather
- ☐ Antenna position marked on a map
- ☐ Kit complete, handed over

Before the session ends

- ☐ Example results reproduced
- ☐ Own data converted and validated
- ☐ Validation plots saved
- ☐ Version log started
- ☐ Benchmark CSV checked and uploaded

Before the deadline

- ☐ Final CSV files uploaded
- ☐ Report (PDF) submitted
- ☐ Repository reproduces both CSV files
- ☐ AI-use statement and prompts included
- ☐ Every member can explain the code

No data, no report: back up the raw UBX file before leaving the site

06

Example Dataset: Block X

Section 6 of 7

- What it contains and expected results
- Site, set-up and RTKLIB
- Python WLS-SPP results

1 Lab Plan & Groups
2 Field Data Collection
3 Processing & Own-Data Validation
4 Benchmark Challenge & Leaderboard
5 Report & Checklist
6 **Example Dataset: Block X**
7 Resources

The Example Dataset: Your Warm-Up

Recorded at Block X in 2025 · the following slides walk through it step by step

What you get

Raw .ubx log, RINEX .obs/.nav, the five input CSV files

Receiver solution (NAV-HPPOSECEF.csv)

Example Python WLS-SPP code and its plots

How to use it

Run the code before the lab; compare with the table

Re-run after every change to your code

Different numbers? Check units, time and file paths

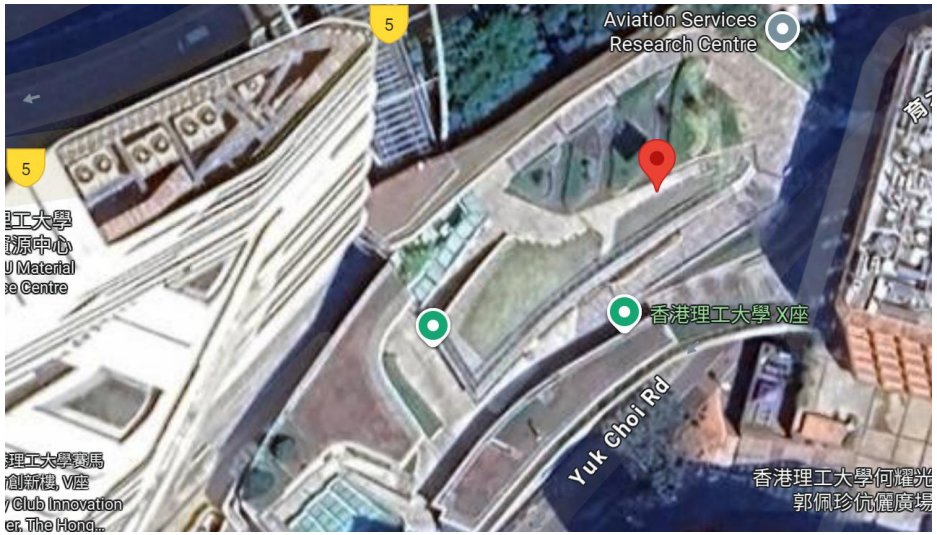
Expected result (GPS only)	2D	3D
Epochs	186	186
Mean error (m)	37.50	49.56
RMS error (m)	37.91	51.04
95th percentile (m)	45.21	73.72
Maximum (m)	51.70	113.73

Errors relative to the receiver solution. In the 2025 plots the receiver solution is labelled "Ground Truth".

If you reproduce these numbers, your environment and code are working

Example: Data Collection Site (Block X)

Experimental Site



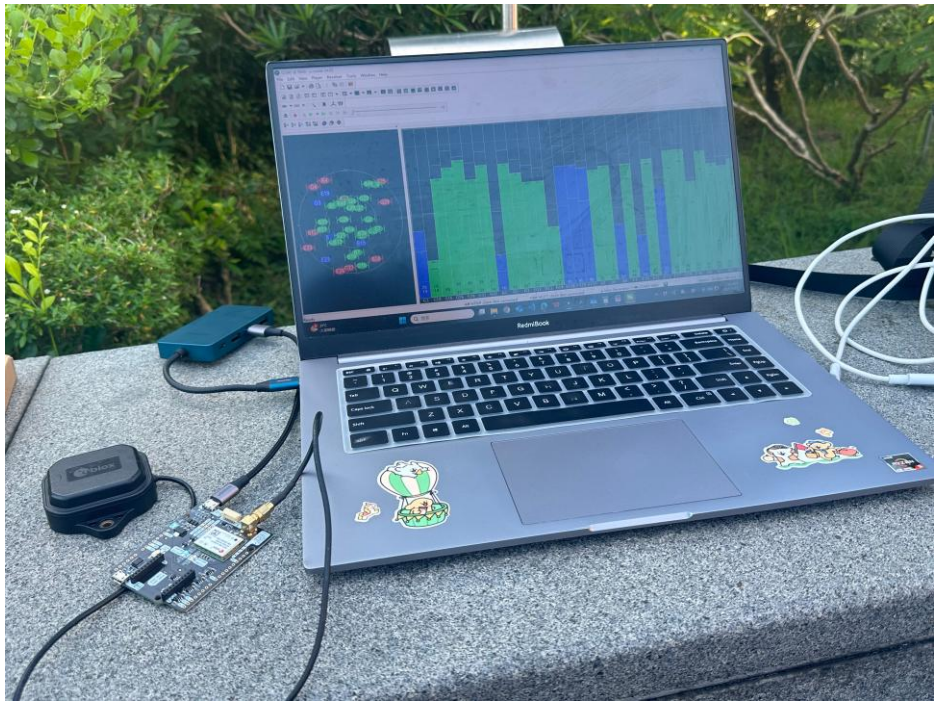
The experiment is to be geographically **restricted to Block X, The Hong Kong Polytechnic University**

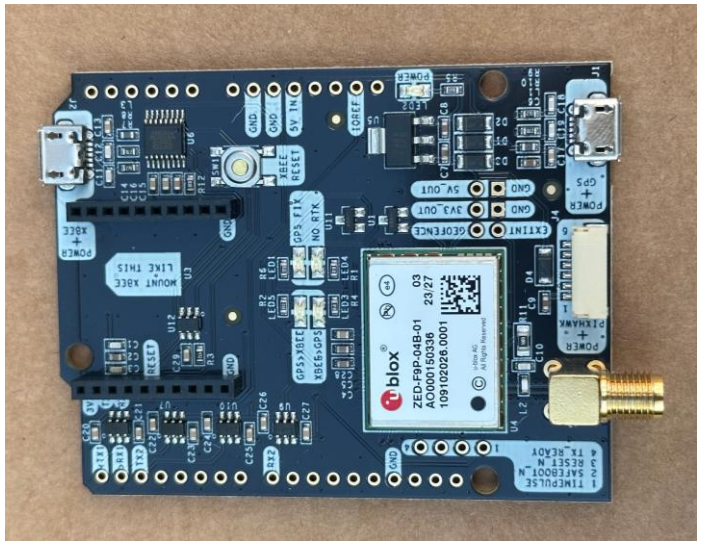
Experimental Environment



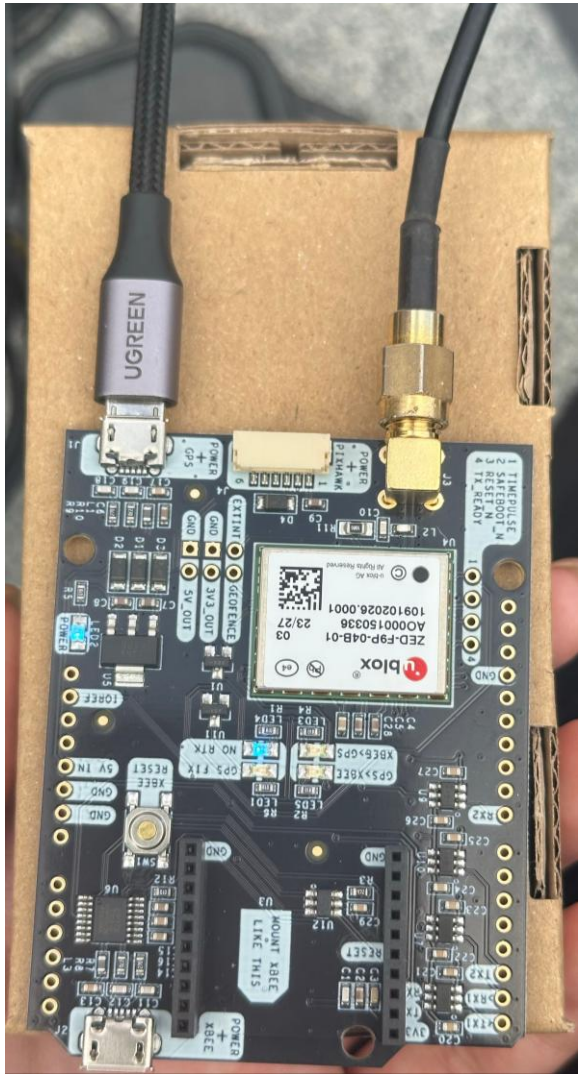


Example: Equipment Set-up









The equipment used for data collection:

- One laptop
- U-blox ZED-F9P + Antenna / U-blox boards

Device connection details

Example: Data Collection



Example: RTKLIB Processing

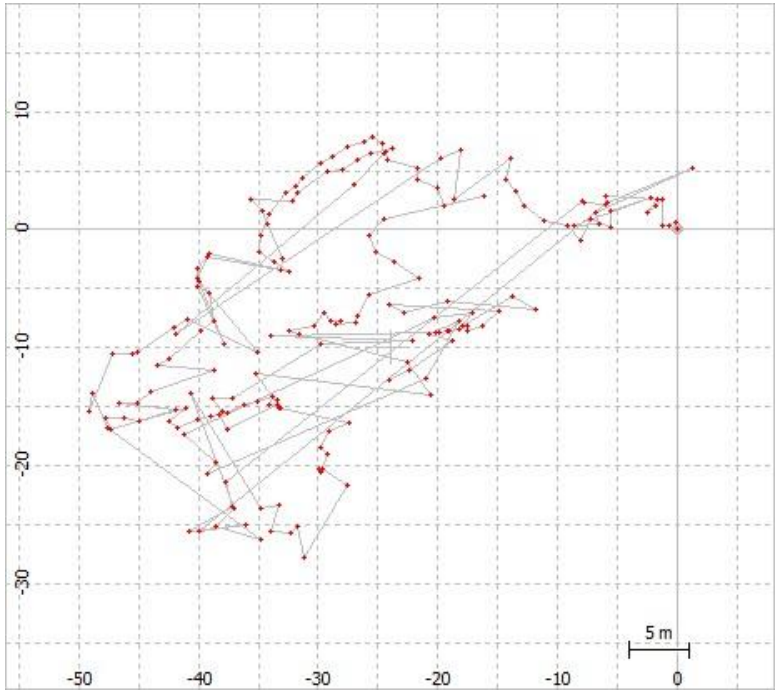
COM6___9600_250916_020025_blockx.ubx

COM6___9600_250916_020025_blockz.**obs**
COM6___9600_250916_020025_blockz.**nav**

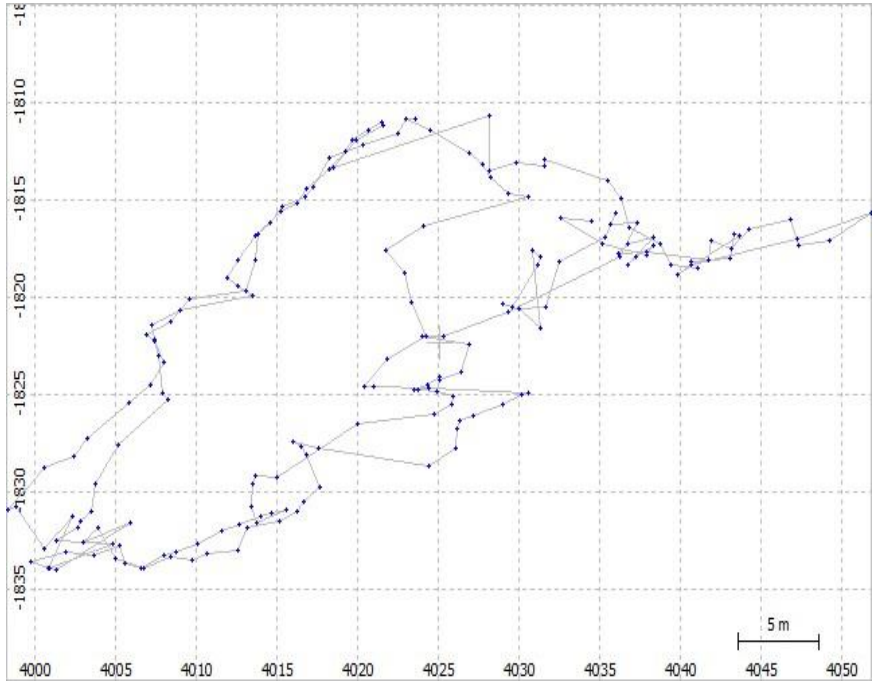
RTKCONV



Receiver reference



RTKPOST:
Single Point Positioning
(SPP)

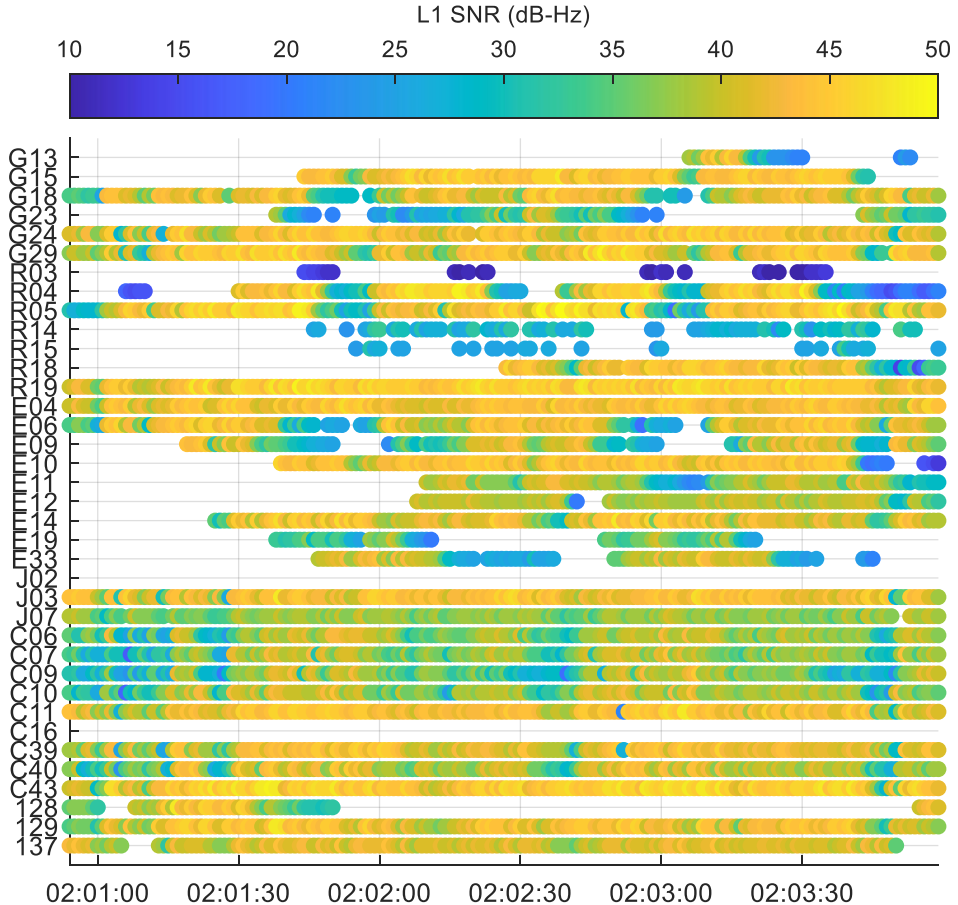


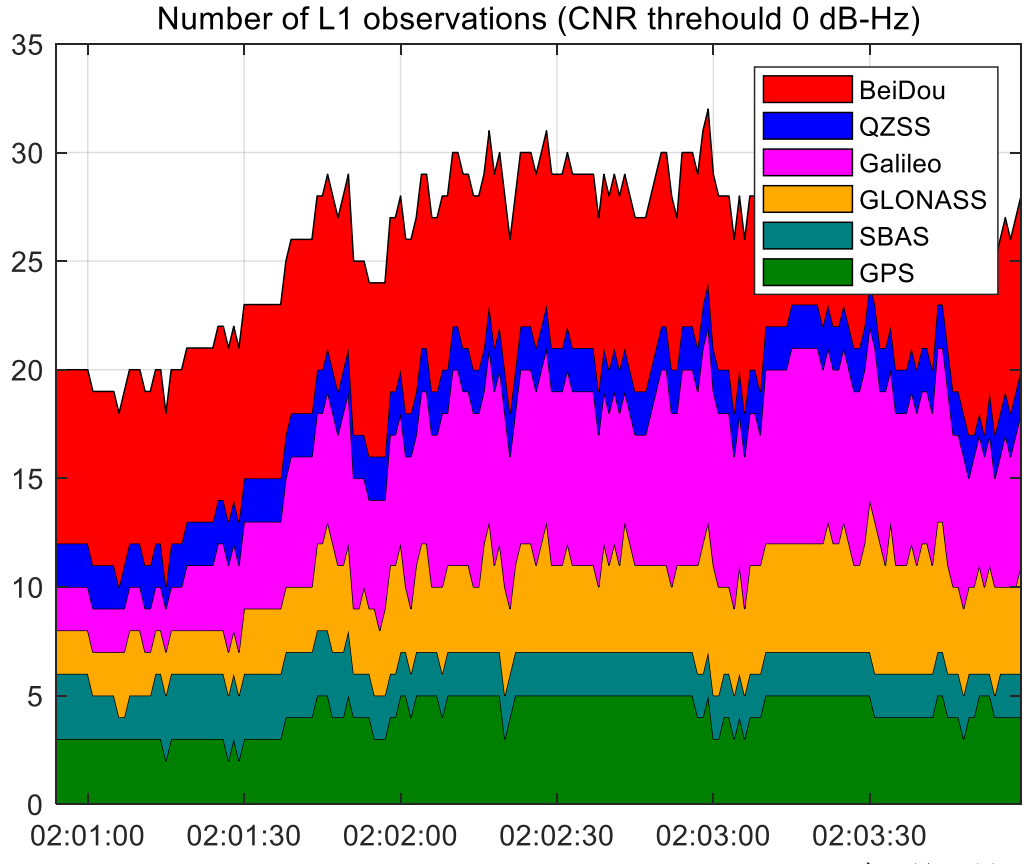
RTKPOST:
DGNSS Positioning

Example: matRTKLIB rinex2csv (1/2)

1. matRTKLIB

```
%% Read RINEX observation/navigation file
obs = gt.Gobs(fullfile('COM6___9600_250916_020025_blockz.obs'));
nav = gt.Gnav(fullfile('COM6___9600_250916_020025_blockz.nav'));
obs.plot;
obs.plotNSat;
obs.plotSky(nav);
```



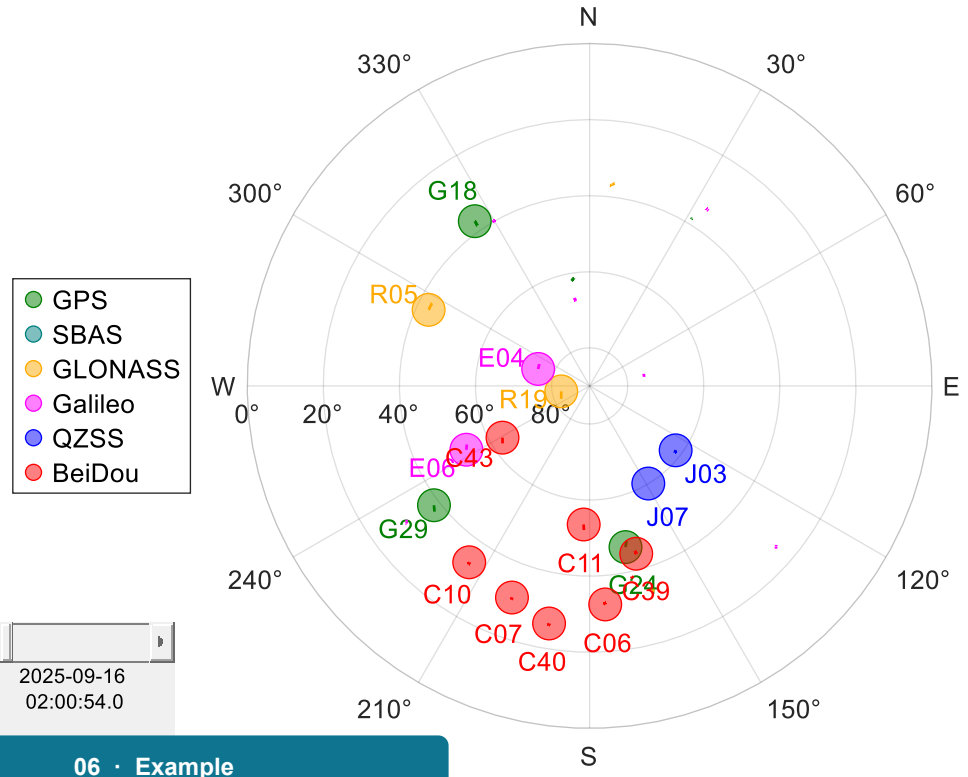


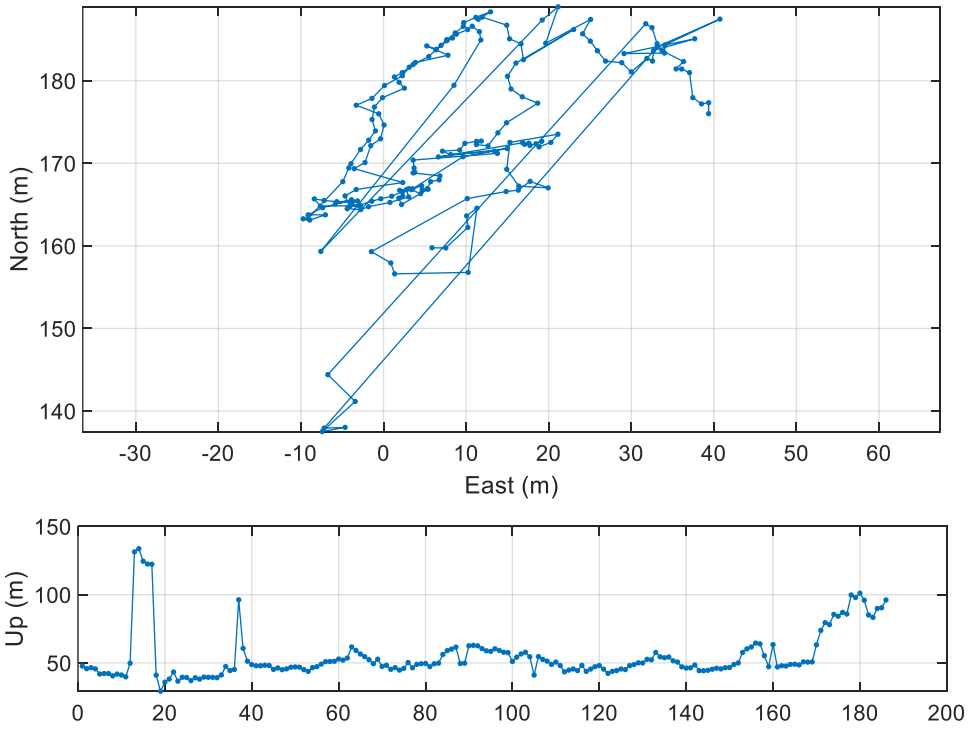
Example: matRTKLIB rinex2csv (2/2)

1. matRTKLIB

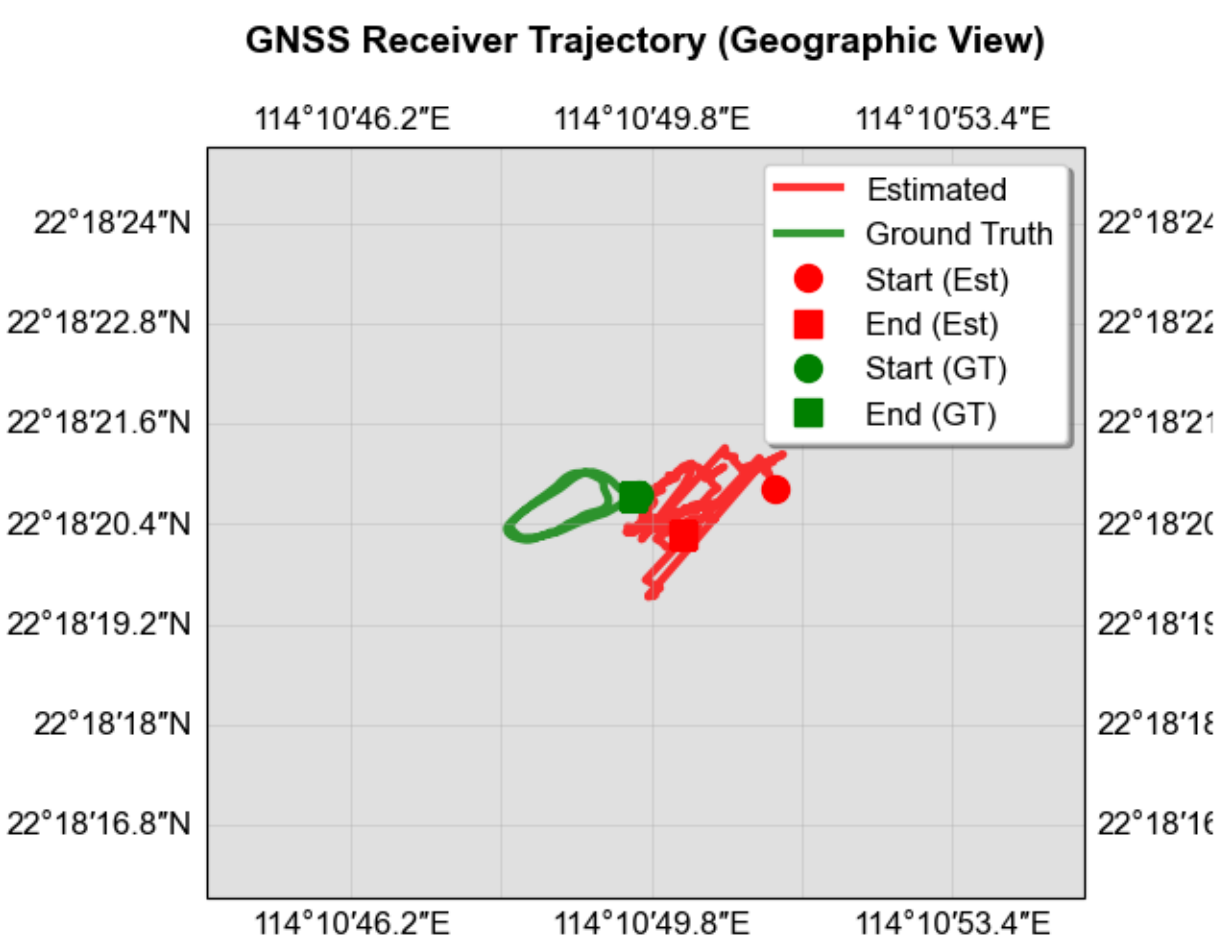
```
%% Read RINEX observation/navigation file
obs = gt.Gobs(fullfile('COM6___9600_250916_020025_blockz.obs'));
nav = gt.Gnav(fullfile('COM6___9600_250916_020025_blockz.nav'));
obs.plot;
obs.plotNSat;
obs.plotSky(nav);
```

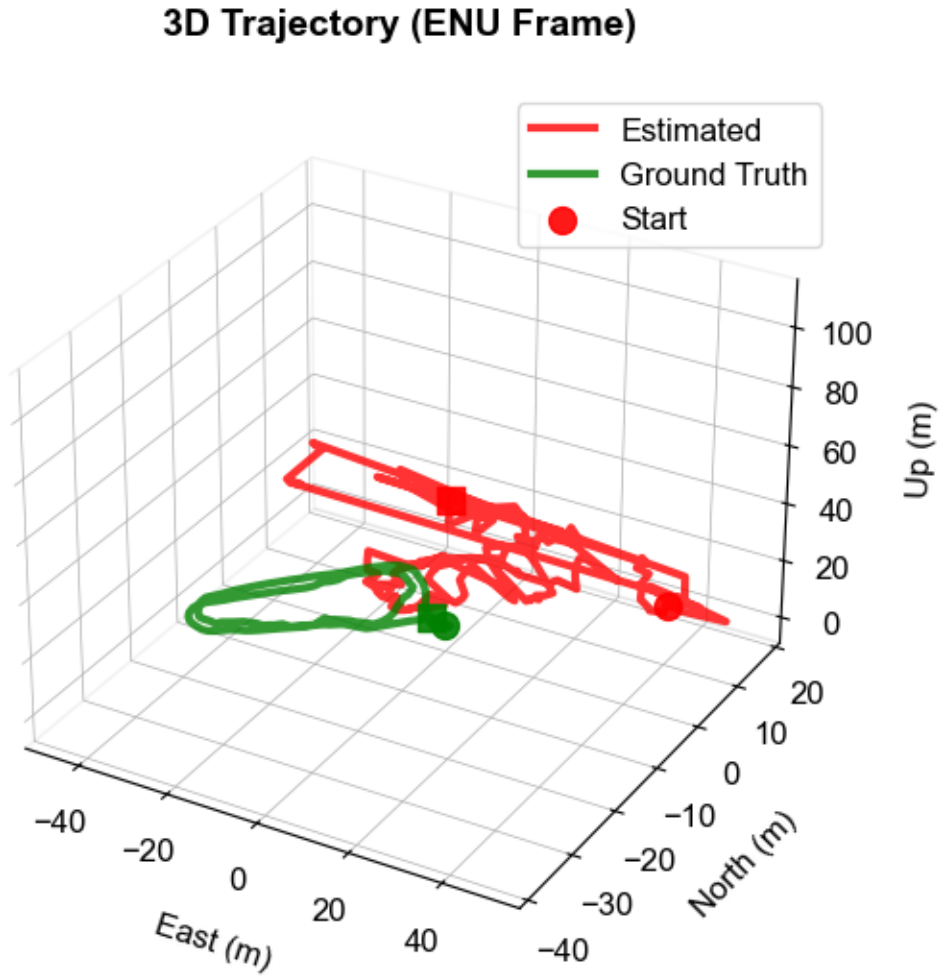
```
% Write the matrices to CSV files
writematrix(pseudoranges_meas_mat, 'pseudoranges_meas.csv');
writematrix(satellite_clock_bias_mat, 'satellite_clock_bias.csv');
writematrix(ionospheric_delay_mat, 'ionospheric_delay.csv');
writematrix(tropospheric_delay_mat, 'tropospheric_delay.csv');
writematrix(satellite_positions_mat, 'satellite_positions.csv');
```



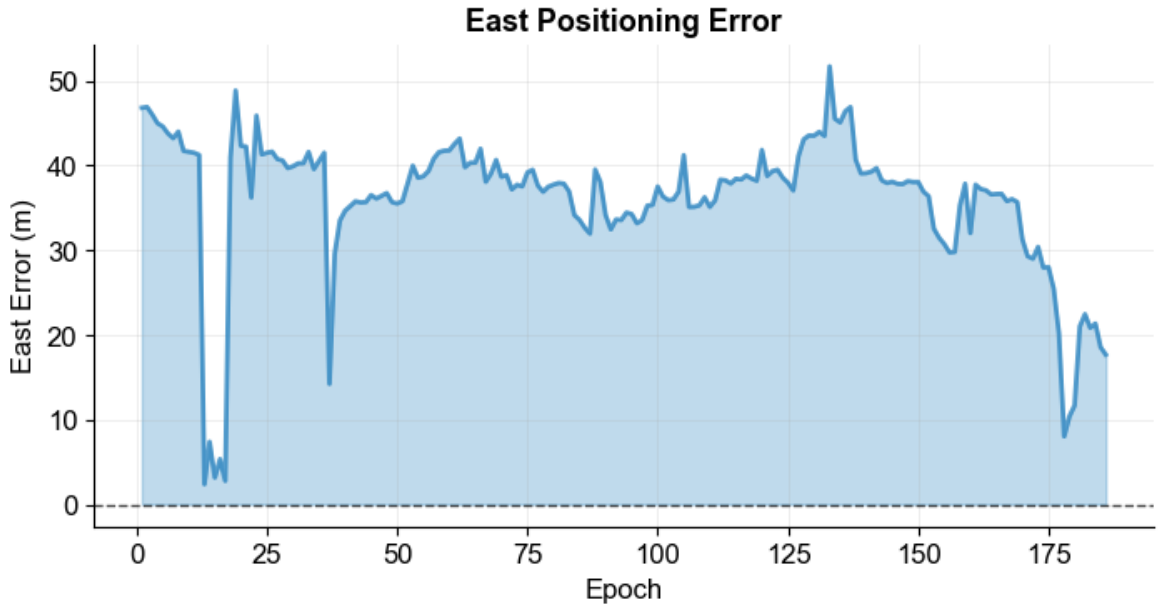


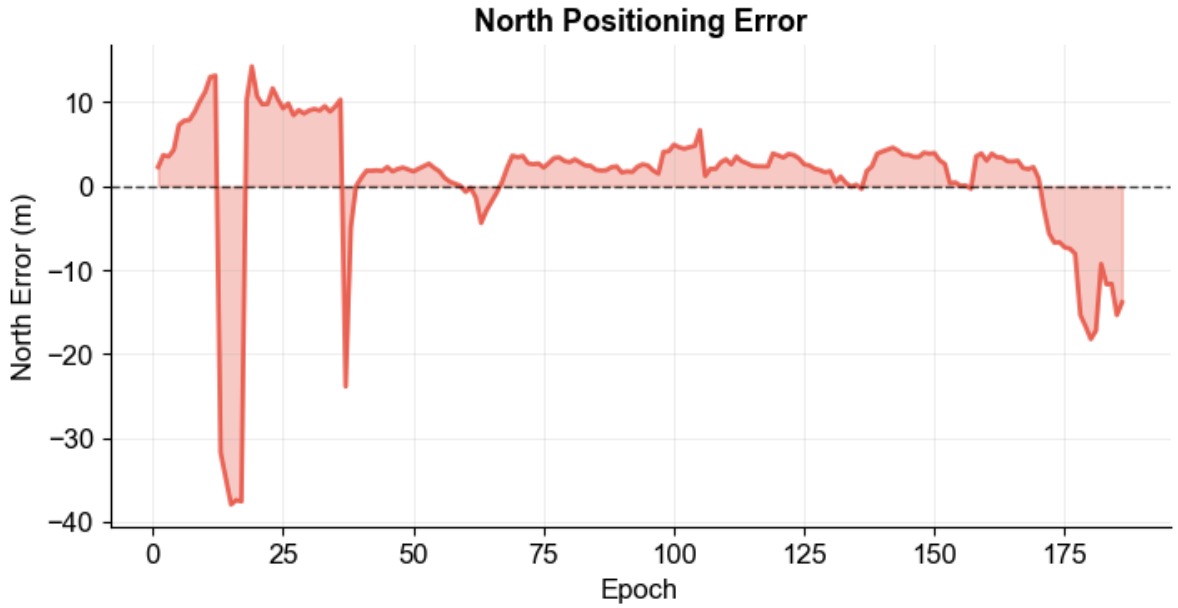
Example: Python WLSE-SPP Results (1/5)

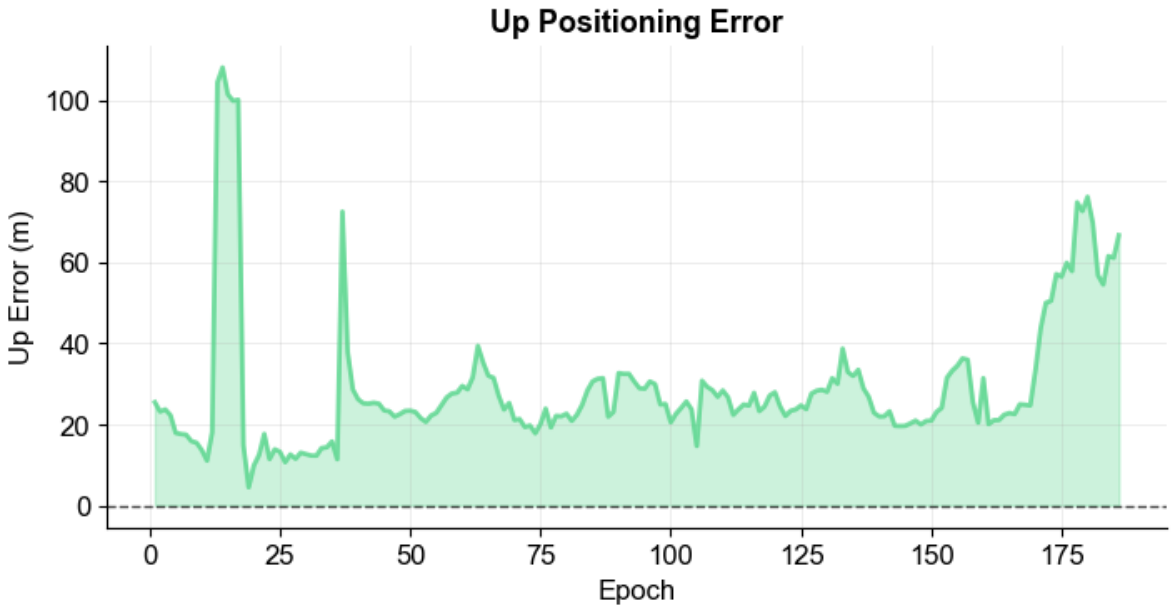


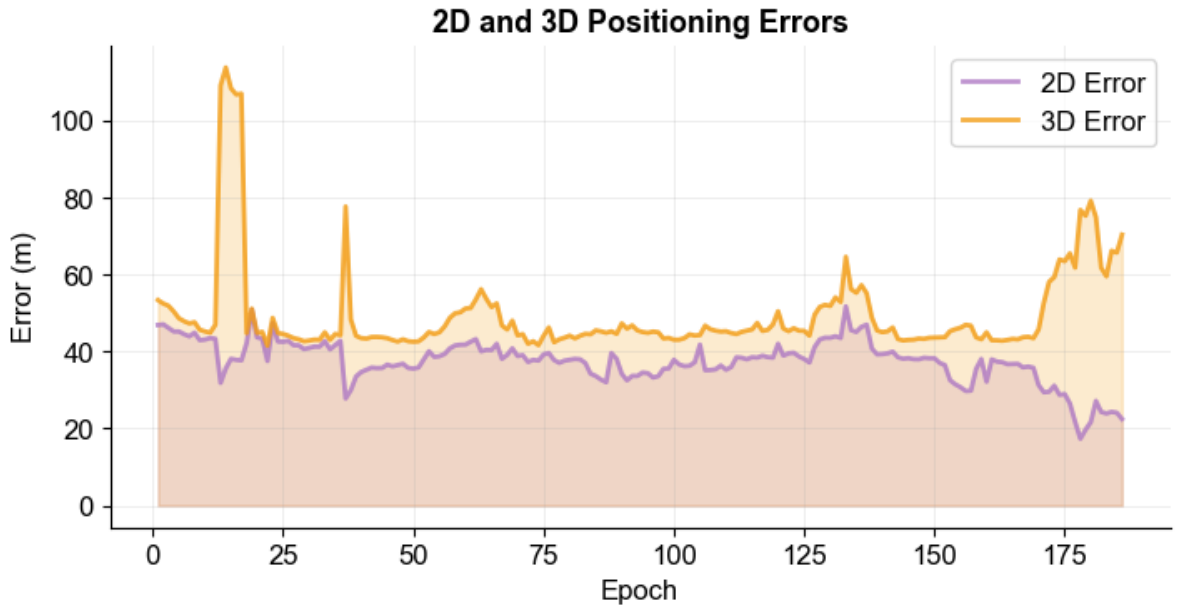


Example: Python WLSE-SPP Results (2/5)

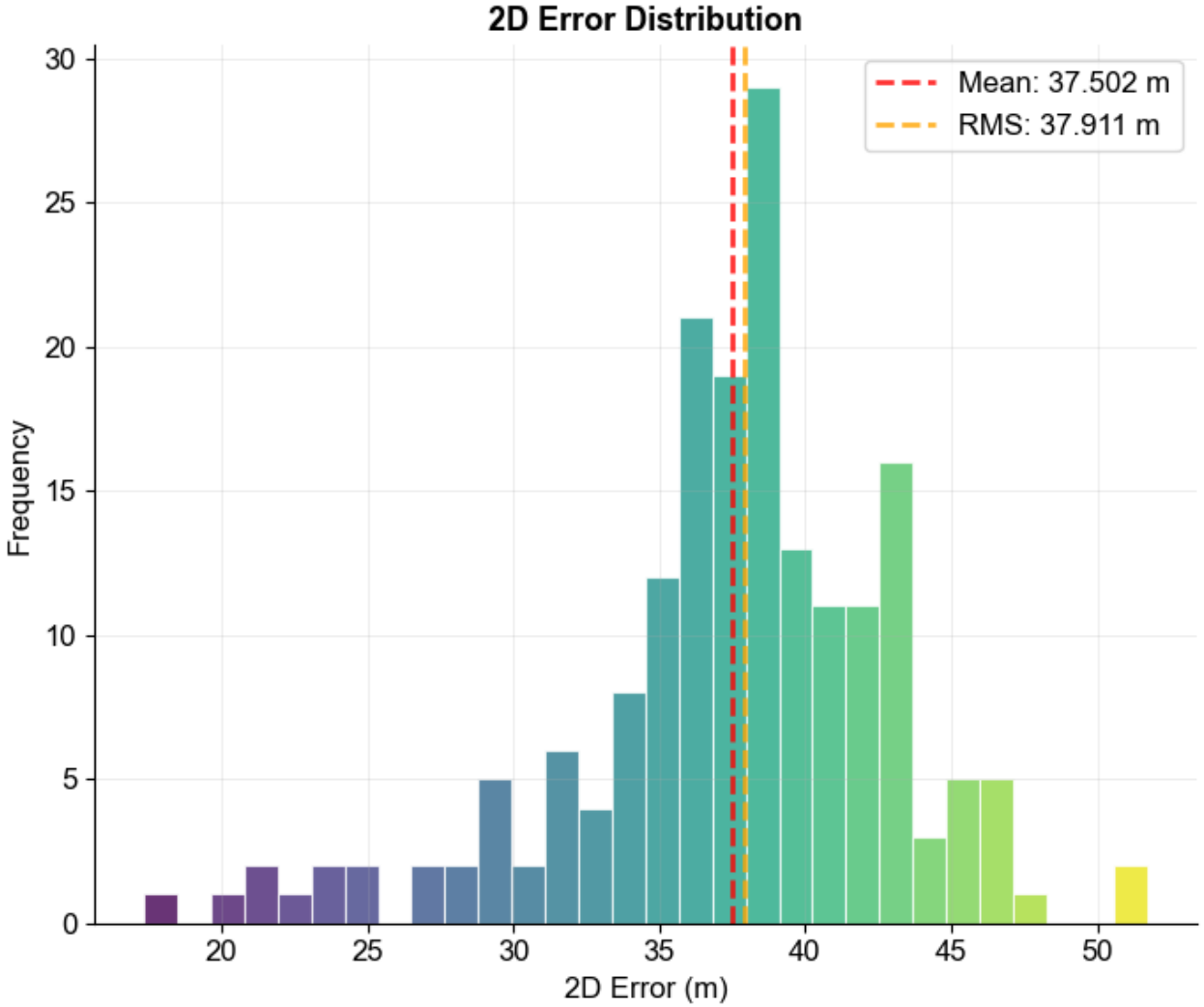


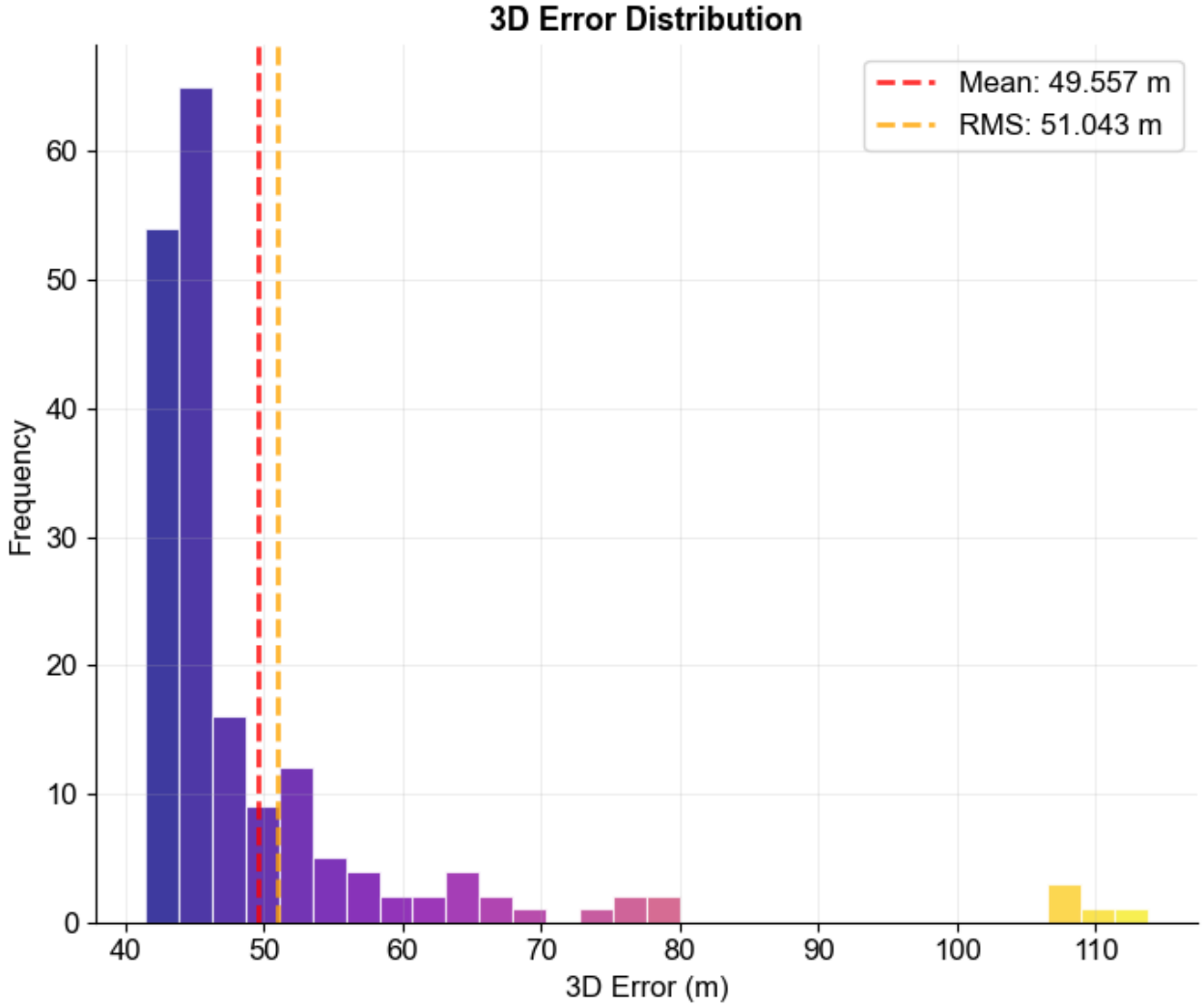




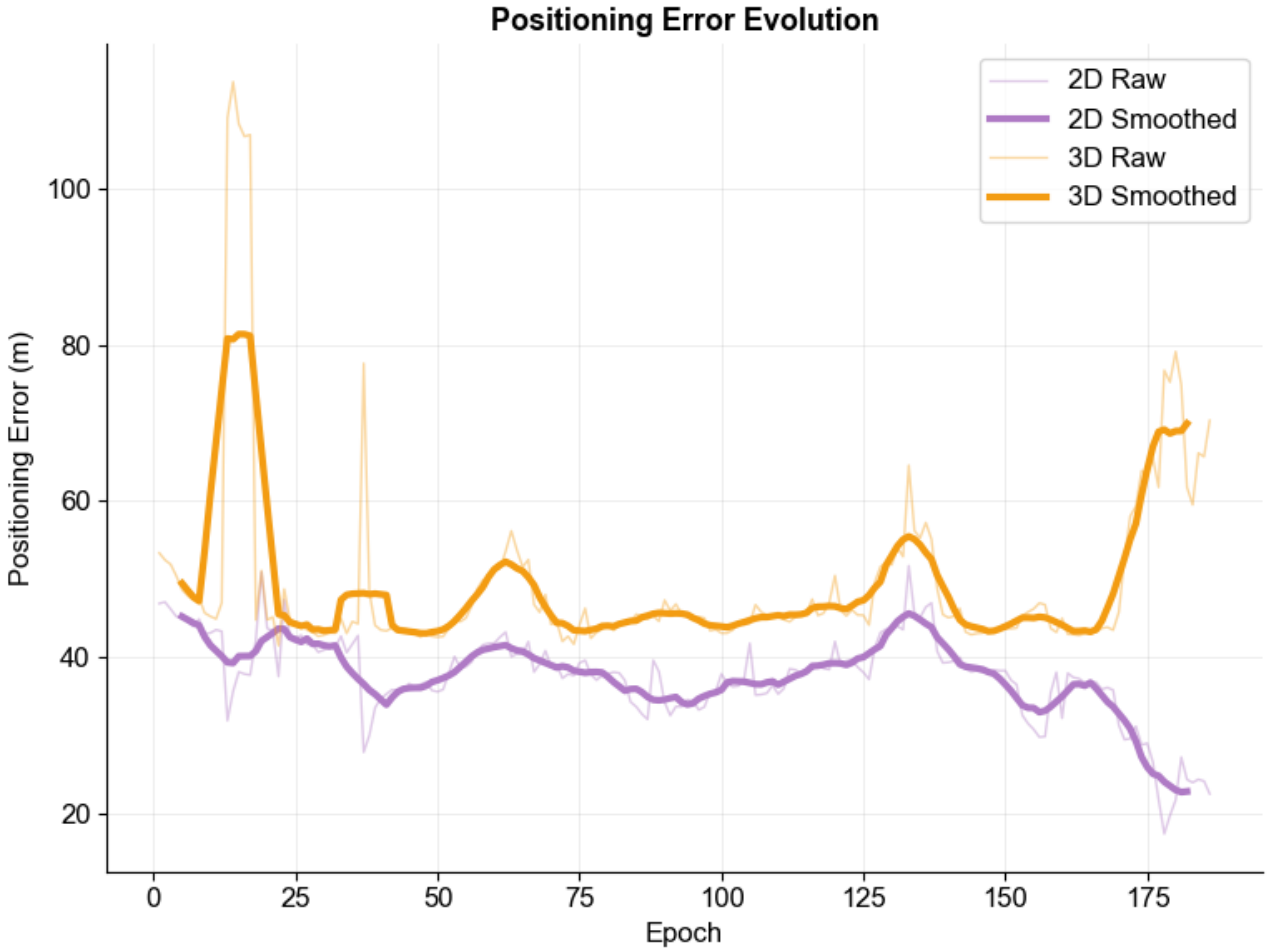


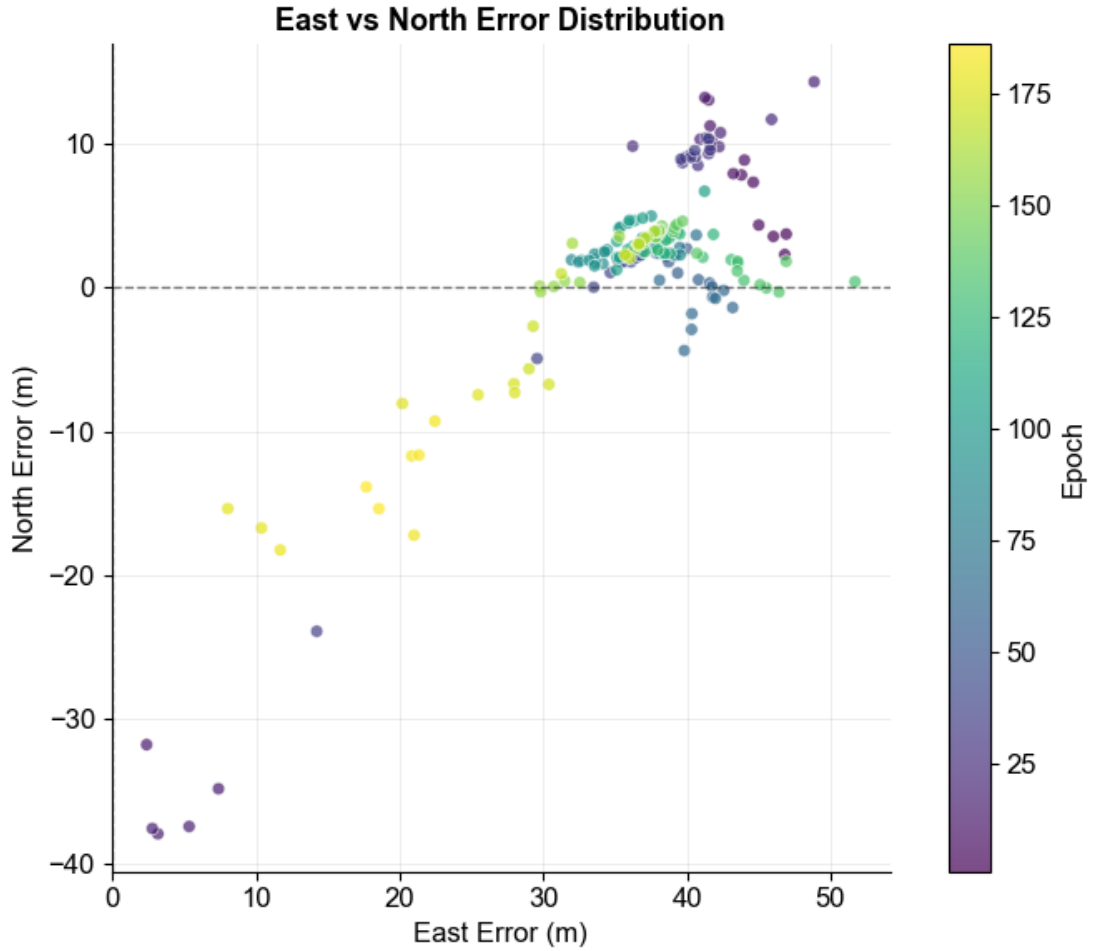
Example: Python WLSE-SPP Results (3/5)





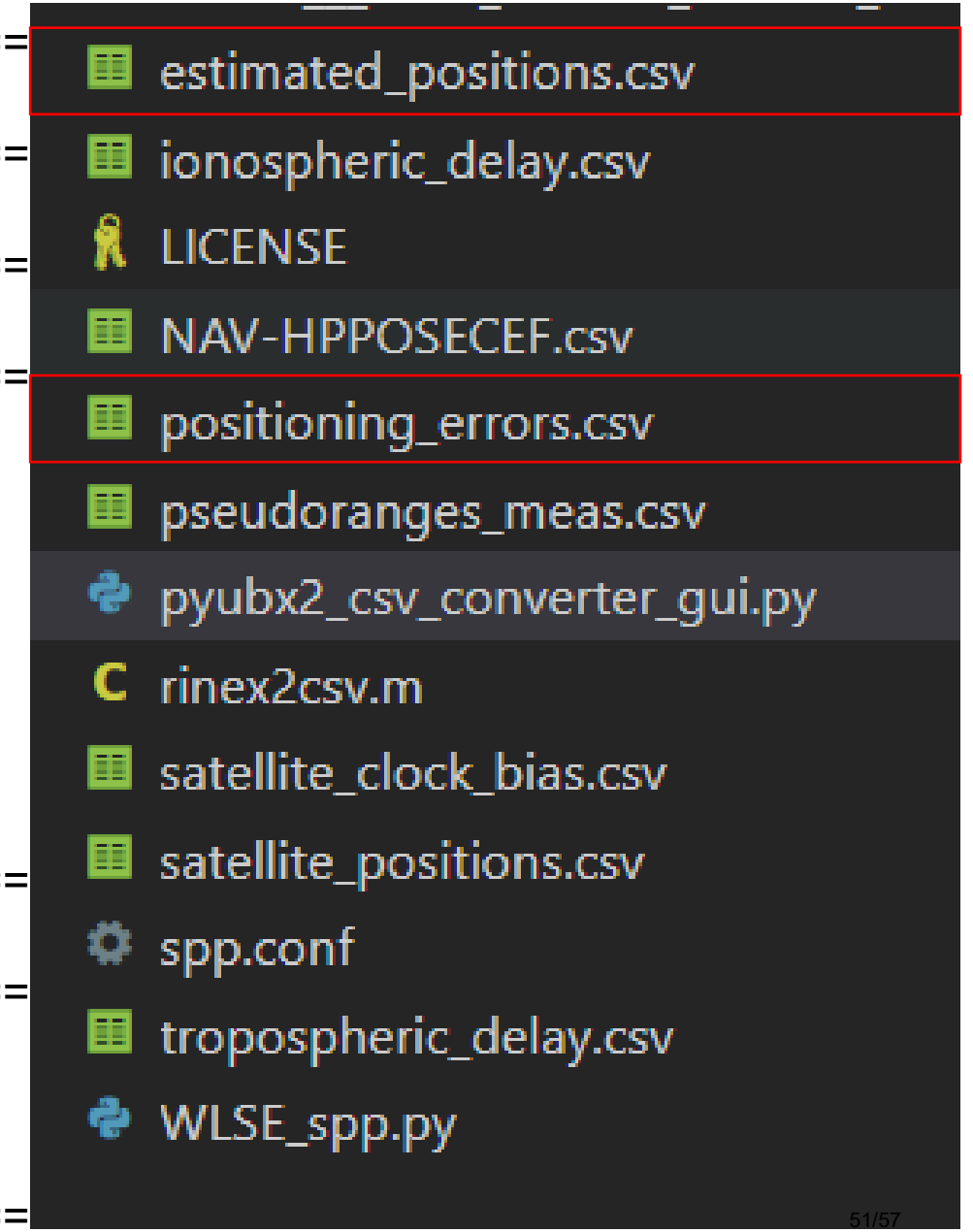
Example: Python WLSE-SPP Results (4/5)





Example: Python WLSE-SPP Results (5/5)

```
================================================================
        GNSS SPP POSITIONING RESULTS SUMMARY
================================================================
Total Epochs Processed: 186
================================================================
        POSITIONING ACCURACY STATISTICS
================================================================
Metric              2D Error (m)    3D Error (m)
----------------------------------------------------
Mean                 37.502          49.557
RMS                  37.911          51.043
Standard Deviation    5.553          12.227
Maximum              51.699         113.729
Minimum              17.350          41.472
95th Percentile      45.208          73.716
================================================================
              COORDINATE RANGES
================================================================
Latitude Range:  22.305424°  to 22.305920°
Longitude Range: 114.180416°  to 114.180938°
Altitude Range:  10.550 m to 114.610 m
================================================================
```



07

Resources

Section 7 of 7

- Lab tutorial
- Software documentation
- Further reading

1 Lab Plan & Groups
2 Field Data Collection
3 Processing & Own-Data Validation
4 Benchmark Challenge & Leaderboard
5 Report & Checklist
6 Example Dataset: Block X
7 **Resources**

Resources and References

[1] AAE4203 Laboratory Tutorial 2026/27 (Word/PDF) — full instructions, theory and worked examples; on Blackboard.

[2] T. Takasu, RTKLIB Manual v2.4.2; rtklibexplorer RTKLIB 2.5 (RTKCONV, RTKPOST, RTKPLOT).

[3] u-blox, u-center User Guide; ZED-F9P Integration Manual (UBX protocol).

[4] L.-T. Hsu, N. Kubo, W. Wen et al., “UrbanNav: an open-sourced multisensory dataset for benchmarking positioning algorithms designed for urban areas,” ION GNSS+ 2021 — the format model for the benchmark dataset.

[5] pyubx2 (UBX parsing) and MatRTKLIB (rinex2csv.m) — links in the tutorial.

[6] Lands Department, HKSAR, SatRef RINEX data, www.geodetic.gov.hk.

[7] P. Misra and P. Enge, Global Positioning System: Signals, Measurements, and Performance, 2nd ed., 2011.

[8] AAE4203 course GitHub repository 2026/27 (code and tools): github.com/Sledge-hammer-Yin/Lab1

Thank You

Dr Weisong Wen · Assistant Professor
Trustworthy AI and Autonomous Systems Laboratory (TAS Lab)
Department of Aeronautical and Aviation Engineering
The Hong Kong Polytechnic University
Email: welson.wen@polyu.edu.hk · Office: R820 · Phone: 3400 8234

Questions?

If you have any questions, please contact me or the TAs.



TAS Lab, 2026

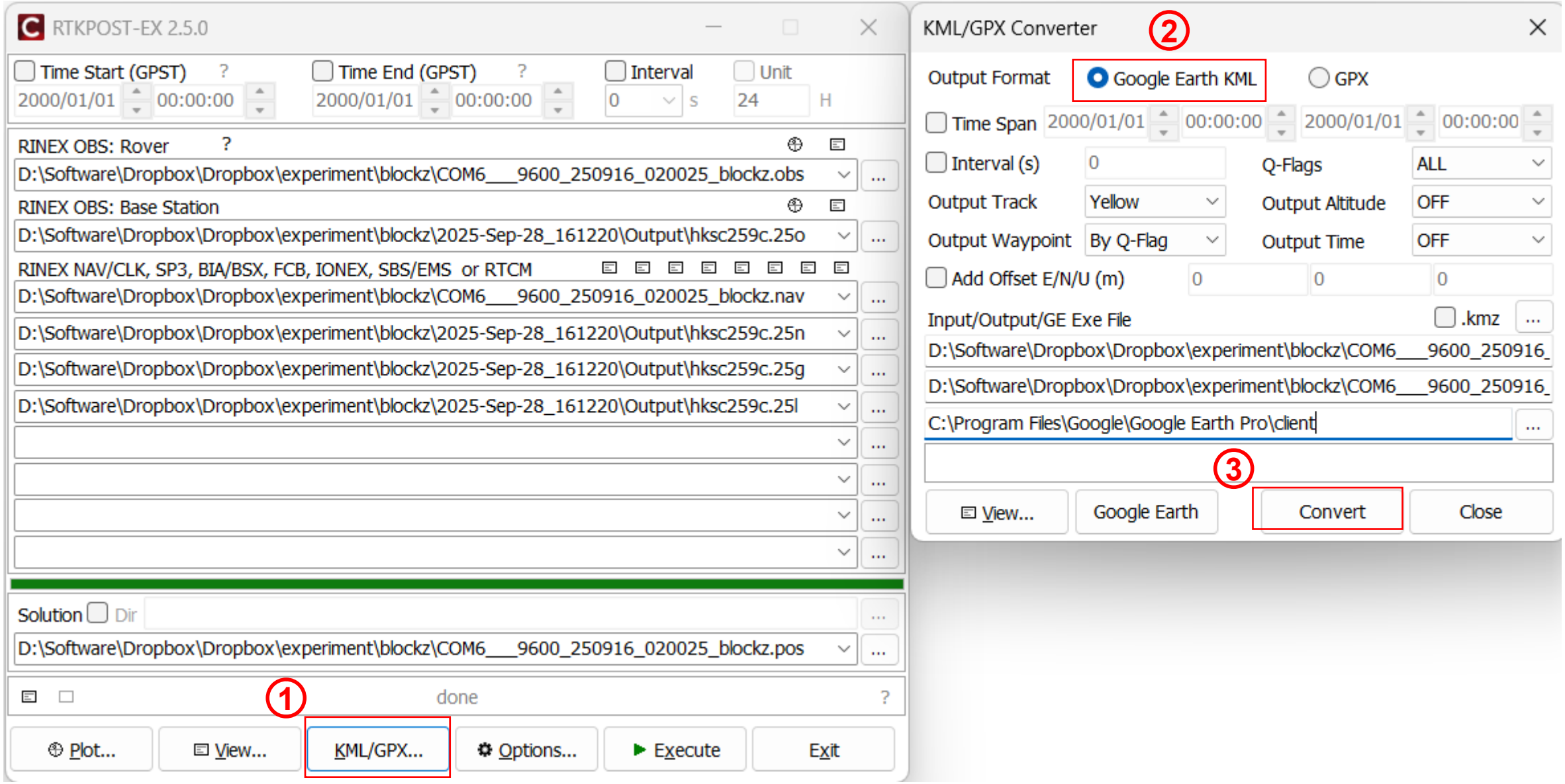
A

Additional RTKLIB Screenshots

Supplementary material

- RTKPOST settings
- RTK vs ground truth

Appendix: RTKPOST Settings (Reference)



Appendix: RTK Solution vs Receiver Reference

